MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
6000 VILLAGE: CLASS-D ACCOMMODATION
DESIGN OF BUILDING STRUCTURE

Document Number **MD-216-6000-EG-CV-CAL-1486**
Revision: **D01**
Revision Date: **31.10.2023**
Status: **AS BUILT**

Outotec Document Number:	900707	GAU01	AN03	CED01	60003	OU100668405	2
L&T Document Number:	O19045	910	C	K	24207	001	2

REVISION HISTORY						
Rev	**Date**	**Revision Text**	**Prepared By**	**Reviewed By**	**Approved By**	**Employer**
A01	28.01.2020	FIRST ISSUE	MF	AVG	SJM	
C01	11.03.2020	ISSUED FOR CONSTRUCTION	MF	AVG	SJM	
D01	31.10.2023	AS BUILT	MF	AVG	SJM	

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document
Typ.	Typical
HCS	Hollow Core Slab
Fc'	Concrete Compressive Strength
Fy	Yield Strength of Reinforcing Steel
ϕ	Strength Reduction Factor
SBC	Saudi Building Code
SASO	Saudi Standards, Metrology and Quality Organization
ACI	American Concrete Institute
ASCE	American Society of Civil Engineers
PCI	Precast/Prestressed Concrete Institute

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of an Accommodation D Building.

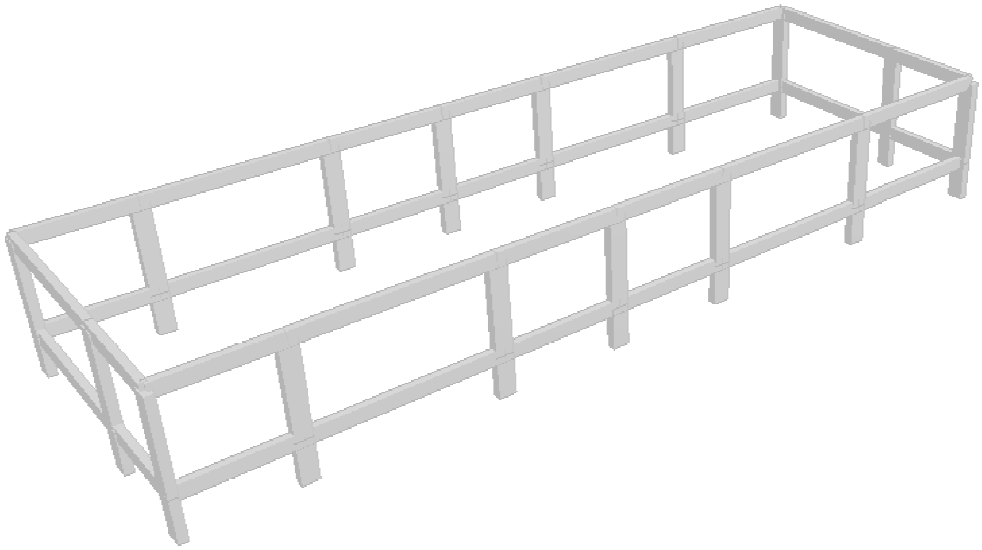
2. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of Accommodation D Building only.

3. DESIGN METHODOLOGY & ASSUMPTIONS

This Building Is a Single-Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have Insulated Block Walls and Cast in Site Framing Beams to support the HSC (Hollow Core Slab)

The Superstructure for this Building was modelled using STAAD software. The Analysis of Frames (Cast-In-Situ Columns & Beams) has been performed meeting design requirements & keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE standards & analysis was done to generate the Column Reactions at the Base that are assumed to be fixed. The Foundation design has been done by importing Column Base Reactions into Foundation Design Software.



STAAD 3D MODEL

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3.1. Codes

[1] SBC-301-2007-Structural Loading & Forces.

[2] SBC-303-2007-Structural Soil & Foundations.

[3] SBC-304-2007-Structural Concrete Structures.

[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 Building Code Requirements.

[8] PCI –MNL-120-04 –Design Handbook, 6th edition.

3.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-CAL-1008-A02.

3.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel

4. DESIGN LOADS

4.1. Dead Load

4.1.1. Roof Dead Loads

- Provision for Electrical Lighting = 0.05 kN/m²

Ceiling

Suspended steel channel system = 0.10 kN/m²

Acoustical Fiber Tile (16mm) = 0.05 kN/m²

Sub-total = 0.2 kN/m²

- Hollow Core Slab

  (HCS-300 Topped) (To be confirmed by Vendor) = **3.88 KN/m2**

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Prestressed Hollowcore Floor

Overall Structural Depth (mm)	Self Wt- KN/M2	No. of Cores in Unit	Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m^2 for Finishes Characteristic Service Loads Kn/m^2							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.00
			Clear Span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = **3.1 kN/m²**

Total Load =3.88+0.2+3.1= 7**.2 kN/m²**

UDL on Edge Ledger Beams (LB1) = 7.2 x 10.33/2=**37.2 Kn/m**

Moment on all the External Beams due to Cantilever Portion of slab and Screed

= (0.15x24+(0.05+0.2)/2x20)*0.6*0.6/2 = **1 knm/m**

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4.1.2. Ground Floor Walls

- 250 thk. Outer CMU (Partially Grouted Cell@800 c-c)

16 x 0.25 x (3.5-0.5) m clear height = **12 kN/m**

4.2. Live Load

4.2.1. Roof Live Loads

Live loads are with reference to Clause-4.2.2, MD-216-1000-EG-ST-CRT-1001

Ordinary Flat (Inaccessible) = **0.5 kN/m²**

Udl Due to Live Load on the Edge Ledger Beam (LB1& LB2) =0.5X10/2= **2.50 kN/m**

Moment on all the External Beams due to Live Load on

Cantilever Portion of slab = 0.5*0.6*0.6/2 = **0.09 kN/m /m**

4.3. Wind Load

Wind loads were automatically generated in STAAD as per ASCE-07. The parameters used for the wind load generation are as follows (Refer STRUCTURAL DESIGN CRITERIA NON-PROCESS BUILDINGS Doc No MD-216-0000-EG-ST-CRT-0001)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

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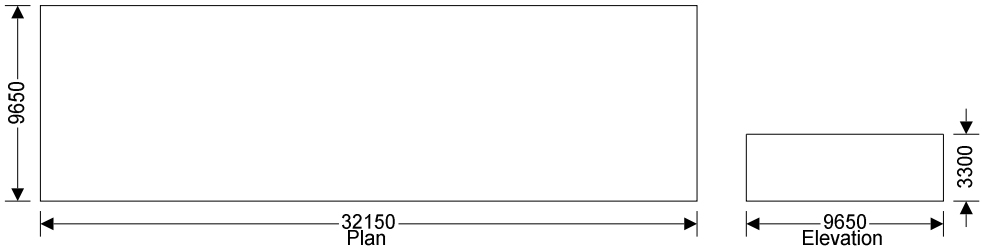
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4.4. Wind Load(Analytical Verification)

WIND LOADING (SBC-301)

Using the directional design method



Building data

Type of roof;	Flat
Length of building;	b = 32150 mm
Width of building;	d = 9650 mm
Height to eaves;	H = 3300 mm
Mean height;	h = 3300 mm

General wind load requirements

Basic wind speed;	V = **49** m/s (or) 177 kmph (Refer Clause-4.2.6, Doc- MD-216-1000-EG-ST- CRT-1001)
Risk category;	II(As per SBC-301, Table 1.6-1)
Velocity pressure exponent coeff;	K_d = 0.85(As per SBC-301, Table 6.4-1)
Exposure category ;	D(As per SBC-301, Table 6.4.2.3)
Enclosure classification ;	Partially enclosed buildings(As per SBC-301, Section-6.2)
Internal pressure coef +ve;	GC_{pi_p} = 0.55(As per SBC-301, Figure-7.2-1)
Internal pressure coef –ve;	GC_{pi_n} = -0.55(As per SBC-301, Figure-7.2-1)
Gust effect factor;	G_f = 0.85(As per SBC-301, Section-7.2.7.1)
Importance Factor;	1(As per SBC-301, Section-9.5)

Topography

Topography factor not significant;	K_{zt} = 1.0(As per SBC-301, Section-6.4.3.2)

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Velocity pressure equation; $q = 0.0473x10^{-3}Kn/m^2 \times K_z \times K_{zt} \times K_d \times V^2 \times I$

V in Km/hr(As per SBC-301, Section-7.2.8)

Velocity pressures table

z (mm)	K_z (SBC-301,Table 7.2-2)	q_z (kN/m^2)
3300	1.03	1.29
3301	1.03	1.29

Peak velocity pressure for internal pressure

Peak velocity pressure – internal (as roof press.); $q_i = 1.289$ kN/m^2

Pressures and forces

Net pressure; $p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$;

Net force; $F_w = p \times A_{ref}$;

Walls load case 1 - Wind 0, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	0.17	106.13	17.78
B	3.30	-0.50	1.29	-1.26	106.10	-133.29
C	3.30	-0.70	1.29	-1.48	31.85	-46.98
D	3.30	-0.70	1.29	-1.48	31.85	-46.98

Overall loading

Projected vertical plan area of wall; $A_{vert_w_0} = b \times H = 106.10$ m^2

Projected vertical area of roof; $A_{vert_r_0} = 0.00$ m^2

Minimum overall horizontal loading; $F_{w,total_min} = p_{min_w} \times A_{vert_w_0} + p_{min_r} \times A_{vert_r_0} = 81.7$ kN

Leeward net force; $F_l = F_{w,wB} = -133.3$ kN

Windward net force; $F_w = F_{w,wA} = 17.8$ kN

Overall horizontal loading; $F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) = 151.1$ kN

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Walls load case 2 - Wind 0, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	1.58	106.13	168.20
B	3.30	-0.50	1.29	0.16	106.10	17.09
C	3.30	-0.70	1.29	-0.06	31.85	-1.85
D	3.30	-0.70	1.29	-0.06	31.85	-1.85

Overall loading

Projected vertical plan area of wall; $A_{vert_w_0} = b \times H = 106.10\ m^2$

Minimum overall horizontal loading; $F_{w,total_min} = p_{min_w} \times A_{vert_w_0} + p_{min_r} \times A_{vert_r_0} = 81.7$ kN

Leeward net force; $F_l = F_{w,wB} = 17.1$ kN

Windward net force; $F_w = F_{w,wA} = 168.2$ kN

Overall horizontal loading; $F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) = 151.1$ kN

Walls load case 3 - Wind 90, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	0.17	31.85	5.34
B	3.30	-0.23	1.29	-0.96	31.85	-30.71
C	3.30	-0.70	1.29	-1.48	106.10	-156.53
D	3.30	-0.70	1.29	-1.48	106.10	-156.53

Overall loading

Projected vertical plan area of wall; $A_{vert_w_90} = d \times H = 31.85\ m^2$

Projected vertical area of roof; $A_{vert_r_90} = 0.00\ m^2$

Minimum overall horizontal loading; $F_{w,total_min} = p_{min_w} \times A_{vert_w_90} + p_{min_r} \times A_{vert_r_90} = 24.5$ kN

Leeward net force; $F_l = F_{w,wB} = -30.7$ kN

Windward net force; $F_w = F_{w,wA} = 5.3$ kN

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Overall horizontal loading; $F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) = 36.0$ kN

Walls load case 4 - Wind 90, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	1.58	31.85	50.49
B	3.30	-0.23	1.29	0.45	31.85	14.43
C	3.30	-0.70	1.29	-0.06	106.10	-6.15
D	3.30	-0.70	1.29	-0.06	106.10	-6.15

Overall loading

Projected vertical plan area of wall; $A_{vert_w_90} = d \times H = 31.85\ m^2$

Projected vertical area of roof; $A_{vert_r_90} = 0.00\ m^2$

Minimum overall horizontal loading; $F_{w,total_min} = p_{min_w} \times A_{vert_w_90} + p_{min_r} \times A_{vert_r_90} = 24.5$ kN

Leeward net force; $F_l = F_{w,wB} = 14.4$ kN

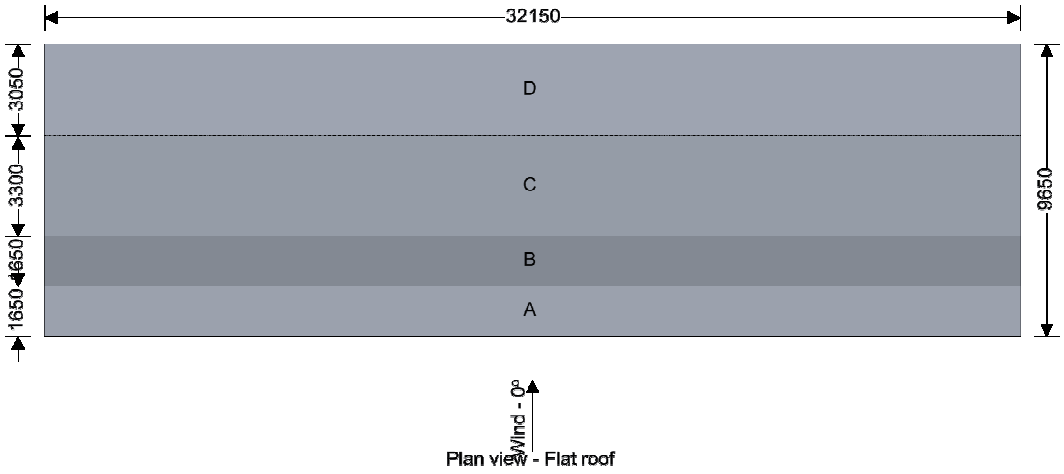
Windward net force; $F_w = F_{w,wA} = 50.5$ kN

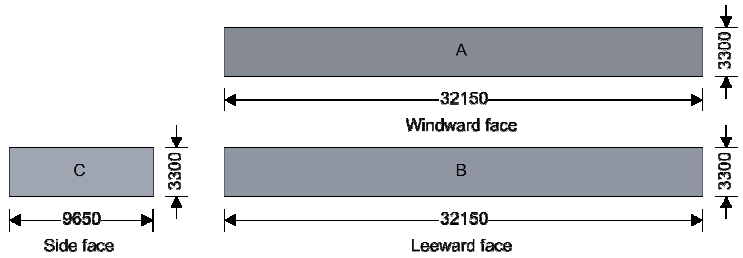
Overall horizontal loading; $F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) = 36.1$ kN

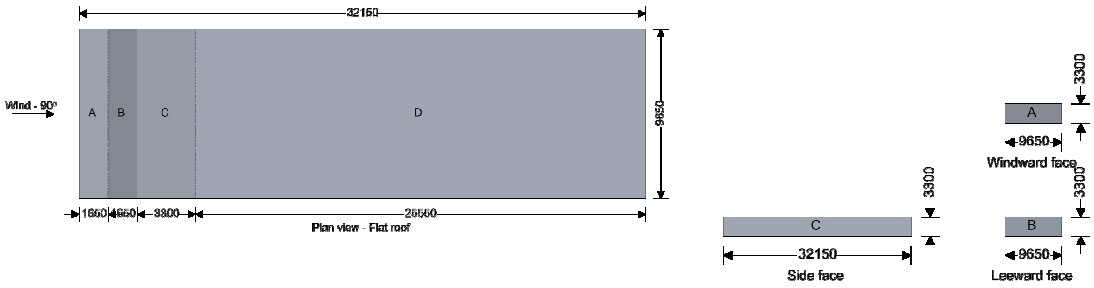
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So from the above analysis, the Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

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4.5. Seismic Load

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

S_S = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 Ref IBC Section 1613.5.4

Where S_{D_1} Damped Design Spectral Response Acceleration

Site Class = D

Site Coefficient, Fa = 1.60(As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40(As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

For Ordinary reinforced concrete moment frame,

Response Modification Factor, Rx = 2.5 (As per SBC-301, Table 10.9.3.2)

Response Modification Factor, Rz = 2.5 (As per SBC-301, Table 10.2)

System Over Strength Factor, Omega = 3 (As per SBC-301, Table 10.2)

Deflection Amplification factor, Cd = 2.5 (As per SBC-301, Table 10.2)

4.6. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

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5. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load

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6. CRACKED SECTIONS

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

For Beams:	I = 0.35Ig
For Columns:	I = 0.70Ig
For Walls:	
Un-cracked	I = 0.70Ig
Cracked	I = 0.35Ig
For Slabs	I = 0.25Ig

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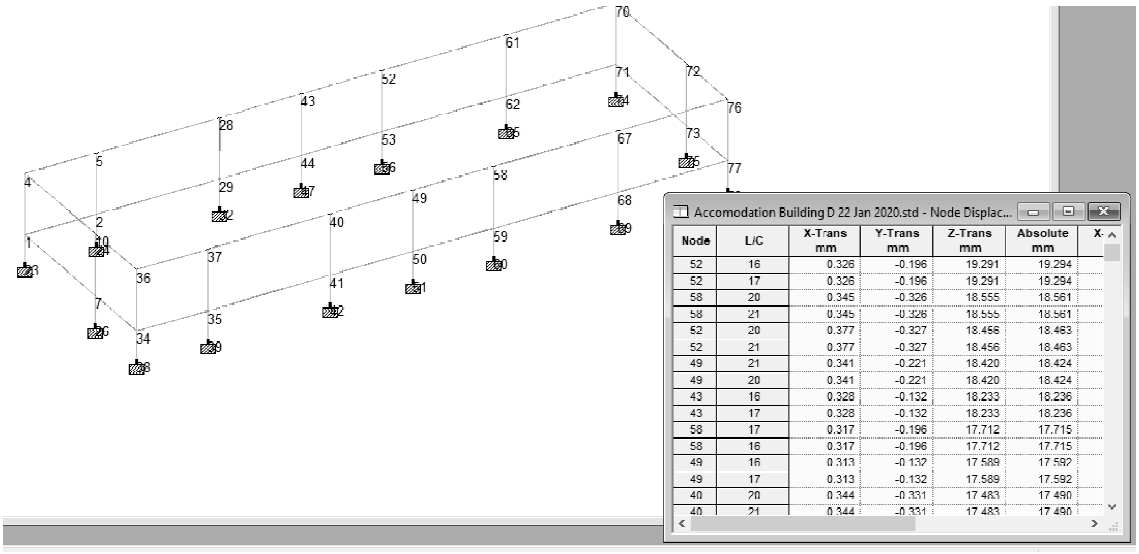
7. DEFLECTIONS/DRIFT

7.1. Maximum Allowable Deflection

L/240 : for member load with live load plus dead load

L/600 : for members supporting masonry under total loads

Where, L = span of member between support



Maximum attained Drift as per Analysis from above figure = 19mm

Allowable Drift as per Table-10.12, SBC-301-2007 is H/100 = 3300/100 = 33mm

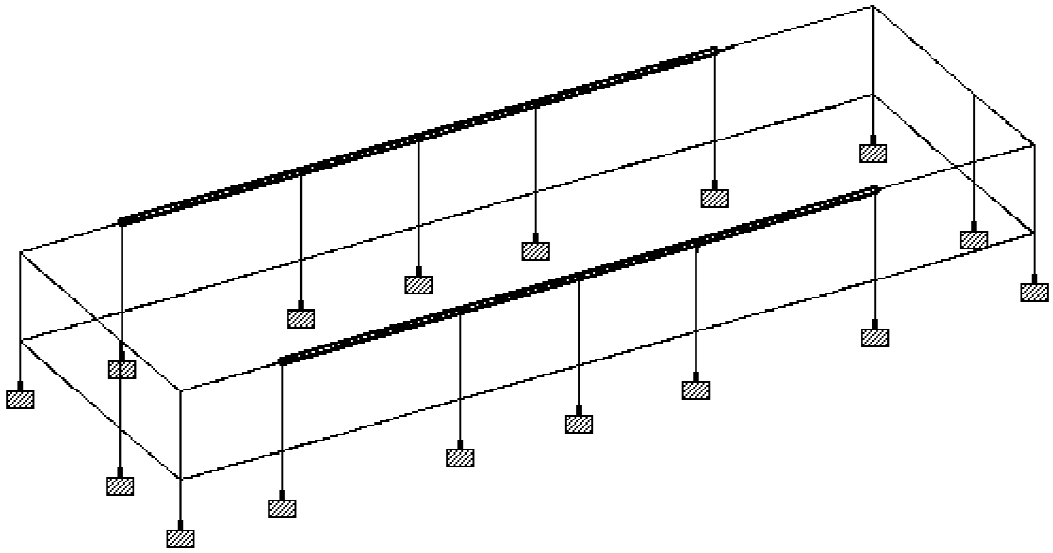
Hence the Structure is safe considering drift check…

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8. CONCRETE ROOF BEAM DESIGN

8.1. Design of Ledger Beam(LB1)



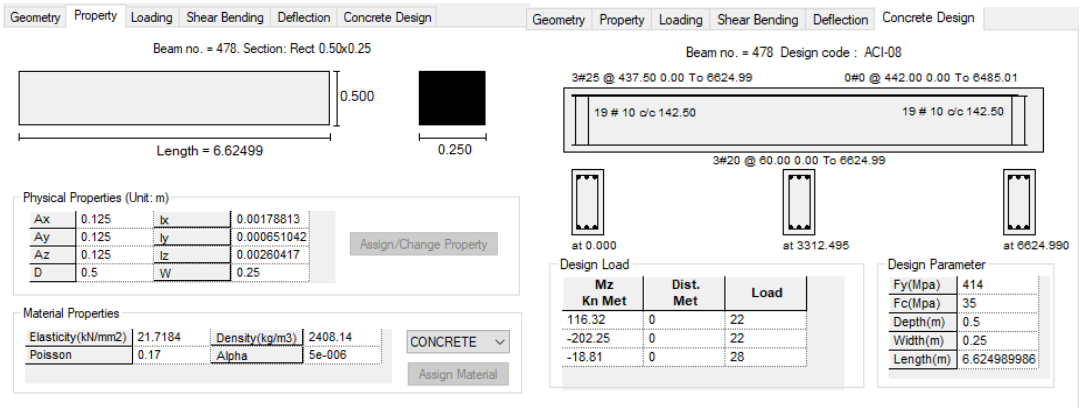
Beam End Summary

All / Summary / Envelope

	Beam	L/C	Node	Fx kN	Fy kN	Fz kN	Mx kNm	My kNm	Mz kNm
Max Fx	557	22 GENERAT	89	524.468	0.878	-11.190	1.288	-5.890	2.212
Min Fx	565	27 GENERAT	62	**-25.642**	51.650	0.071	8.049	-0.115	48.749
Max Fy	503	22 GENERAT	37	46.065	**186.129**	-0.141	-5.682	0.134	194.796
Min Fy	478	22 GENERAT	28	51.218	**-188.683**	0.136	-3.833	0.761	200.297
Max Fz	502	23 GENERAT	35	384.488	0.811	**146.888**	-0.329	-75.935	-3.456
Min Fz	582	23 GENERAT	77	265.433	46.443	**-177.745**	-2.917	78.629	31.113
Max Mx	498	27 GENERAT	34	109.390	-8.537	-14.151	**19.821**	30.665	-10.186
Min Mx	497	27 GENERAT	36	22.203	72.006	-8.727	**-18.155**	15.194	15.071
Max My	502	23 GENERAT	39	392.139	0.811	146.888	-0.329	**144.398**	-4.673
Min My	582	23 GENERAT	78	273.084	46.443	-177.745	-2.917	**-187.989**	-38.551
Max Mz	545	22 GENERAT	61	52.688	-188.318	-0.047	-5.286	0.233	**205.892**
Min Mz	538	31 GENERAT	56	295.980	45.668	-0.454	1.164	-4.472	**-127.130**

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Design of Ledger Beam (LB1)

Maximum Moment required from Analysis = 202 **Knm (Hogging)**

Therefore Ast.req = $$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ =0.9

Ast1.req = **1339 mm^2**

Ast.min= (As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f_c'}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **406mm^2**

Maximum Moment required from Analysis = **116 Knm (Sagging)**

Ast2.req = **742 mm^2**

Hence Provide 3-25 dia-Full Straight Bars Top & 3-20 dia-Full Straight Bottom

Ast1.pro= **1470mm^2 >>>>>>(Ast.min, Ast1.req)**

Ast2.pro= **942mm^2 >>>>>>(Ast.min, Ast2.req)**

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8.1.1. Staad Output For LB1

```
        BEAM  NO.   478 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN -   6625. MM   FY -   414.   FC -   35.   MPA, SIZE -   250. X   500. MMS

LEVEL     HEIGHT      BAR INFO        FROM          TO          ANCHOR
            (MM)                        (MM)          (MM)      STA   END
_____________________________________________________________________________


  1          60.      3 - 20MM          0.          6625.        YES   YES
  2         438.      3 - 25MM          0.          6625.        YES   YES


        B E A M  N O.   478 D E S I G N  R E S U L T S - SHEAR


 AT START SUPPORT - Vu=   133.67 KNS   Vc=   124.19 KNS   Vs=    54.03 KNS
 Tu=     4.22 KN-MET   Tc=     4.2 KN-MET   Ts=     5.6 KN-MET   LOAD     22
 STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2432. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    1.39 SQ.CM.

 AT END   SUPPORT - Vu=   127.00 KNS   Vc=   118.52 KNS   Vs=    50.82 KNS
 Tu=    10.10 KN-MET   Tc=     4.2 KN-MET   Ts=    13.5 KN-MET   LOAD     27
 STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2432. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    3.33 SQ.CM.

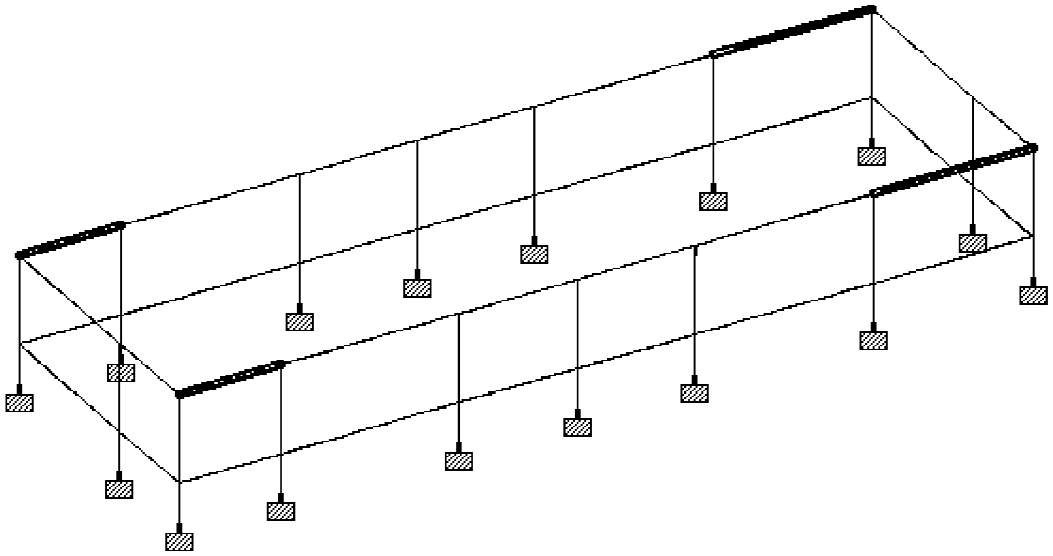
___   5J_____________________   6624X 250X  500_____________________   28J____
|                                                                              |
||==============================================================================||
| 3No25 H 438.| |0.TO 6625| | | | | | | | | | | | | | | | | | | | | | | | | |
| 19*10c/c143 | | | | | | | | | | | | | | | | | | | | | |19*10c/c143| | |
| 3No20 H |60.| |0.TO 6625| | | | | | | | | | | | | | | | | | | | | | | | |
||==============================================================================||
|                                                                              |
|______________________________________________________________________________|

 _______   _______   _______   _______   _______   _______   _______   _______
|       | |       | |       | |       | |       | |       |     |       |
|  ooo  | |  ooo  | |  ooo  | |  ooo  | |  ooo  | |  ooo  |     |  ooo  |
| 3#25  | | 3#25  | | 3#25  | | 3#25  | | 3#25  | | 3#25  |     | 3#25  |
|       | |       | |       | |       | |       | |       |     |       |
| 3#20  | | 3#20  | | 3#20  | | 3#20  | | 3#20  | | 3#20  |     | 3#20  |
|  ooo  | |  ooo  | |  ooo  | |  ooo  | |  ooo  | |  ooo  |     |  ooo  |
|       | |       | |       | |       | |       | |       |     |       |
|       | |       | |       | |       | |       | |       |     |       |
-------------------------------< PAGE 12 Ends Here >-----------------------------
   STAAD SPACE                                                -- PAGE NO.   13
```


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8.2. Design of Ledger Beam(LB2)



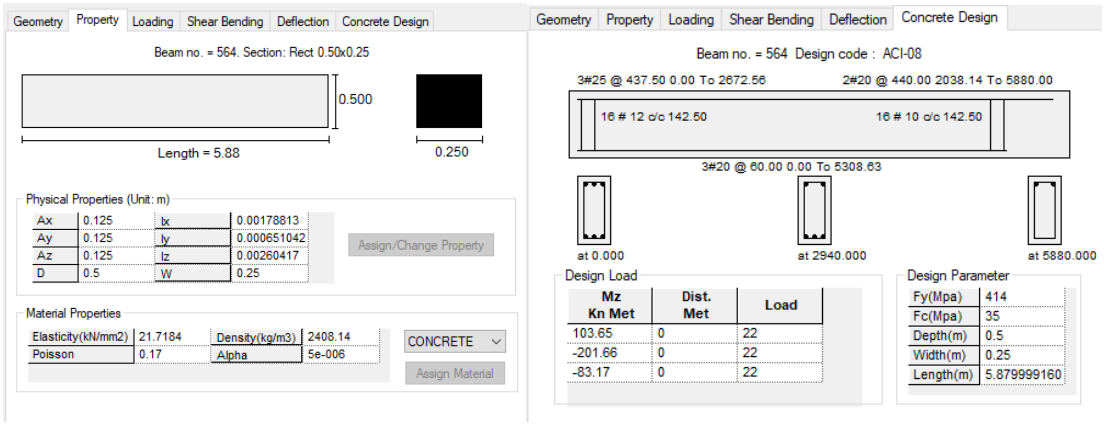
Beam End Summary

All / Summary / Envelope

	Beam	L/C	Node	Fx kN	Fy kN	Fz kN	Mx kNm	My kNm	Mz kNm
Max Fx	557	22 GENERAT	69	624.466	0.679	-11.130	1.288	-5.690	2.212
Min Fx	565	27 GENERAT	62	**-25.642**	51.650	0.071	8.049	-0.115	48.749
Max Fy	503	22 GENERAT	37	46.065	**186.129**	-0.141	-5.682	0.134	194.796
Min Fy	478	22 GENERAT	28	51.218	**-188.683**	0.136	-3.833	0.761	200.297
Max Fz	502	23 GENERAT	35	384.488	0.811	**146.888**	-0.329	-75.935	-3.456
Min Fz	582	23 GENERAT	77	265.433	46.443	**-177.745**	-2.917	78.629	31.113
Max Mx	498	27 GENERAT	34	109.390	-8.537	-14.151	**19.821**	30.665	-10.186
Min Mx	497	27 GENERAT	36	22.203	72.006	-8.727	**-18.155**	15.194	15.071
Max My	502	23 GENERAT	39	392.139	0.811	146.888	-0.329	**144.398**	-4.673
Min My	582	23 GENERAT	78	273.084	46.443	-177.745	-2.917	**-187.989**	-38.551
Max Mz	545	22 GENERAT	61	52.688	-188.318	-0.047	-5.286	0.233	**205.892**
Min Mz	538	31 GENERAT	56	295.980	45.668	-0.454	1.164	-4.472	**-127.130**

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Design of Ledger Beam (LB2)

Maximum Moment required from Analysis = 201 **Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ =0.9

Ast1.req = **1339 mm^2**

Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f_c'}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **406mm^2**

Maximum Moment required from Analysis = **103 Knm (Sagging)**

Ast2.req = **655 mm^2**

Hence Provide 3-25 dia-Full Straight Bars Top & 3-20 dia-Full Straight Bottom

Ast1.pro= **1470mm^2 >>>>>>(Ast.min, Ast1.req)**

Ast2.pro= **942mm^2 >>>>>>(Ast.min, Ast2.req)**

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8.2.1. Staad Output For LB2

```
        BEAM  NO.   564 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN -  5880. MM  FY -  414.  FC -  35.  MPA, SIZE -  250. X  500. MMS

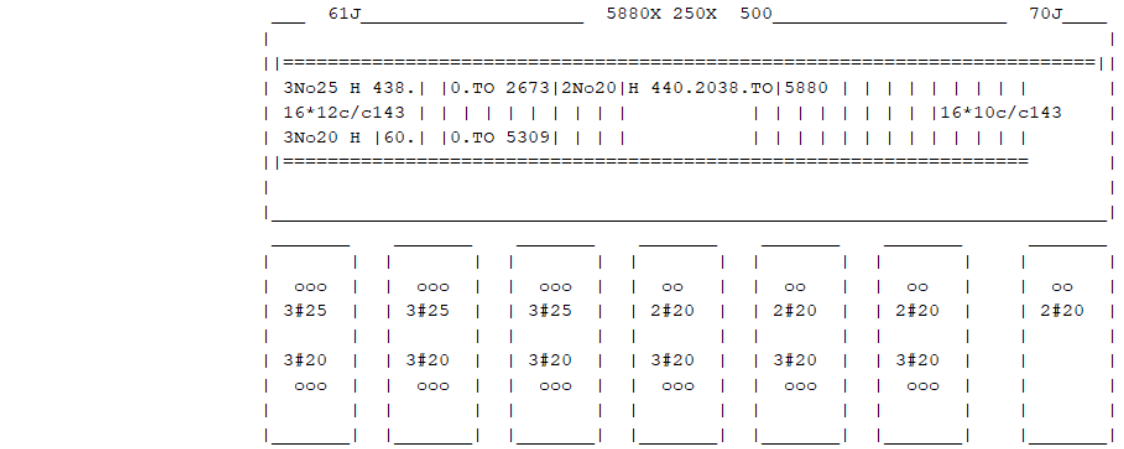
LEVEL     HEIGHT      BAR INFO        FROM          TO          ANCHOR
            (MM)                        (MM)          (MM)       STA  END
_____________________________________________________________________

  1          60.     3 - 20MM          0.          5309.        YES  NO
  2         438.     3 - 25MM          0.          2673.        YES  NO
  3         440.     2 - 20MM       2038.          5880.        NO   YES


        B E A M  N O.   564 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=  121.32 KNS  Vc=  115.31 KNS  Vs=   46.46 KNS
Tu=   16.76 KN-MET  Tc=    4.0 KN-MET  Ts=   22.3 KN-MET  LOAD    27
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2060. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   5.52 SQ.CM

AT END   SUPPORT - Vu=   62.77 KNS  Vc=  140.77 KNS  Vs=    0.00 KNS
Tu=   12.90 KN-MET  Tc=    4.0 KN-MET  Ts=   17.2 KN-MET  LOAD    31
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
```

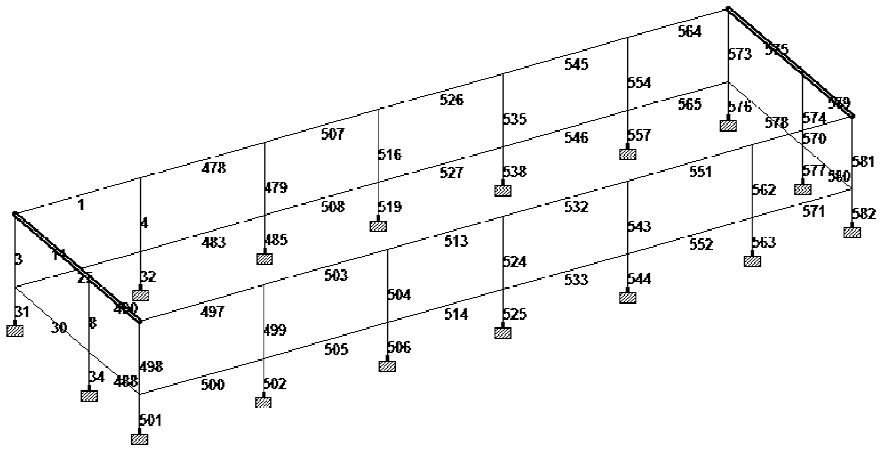


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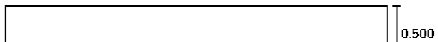
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8.3. Design of Beam-(B1)



Geometry Property Loading Shear Bending Deflection Concrete Design

Beam no. = 11. Section: Rect 0.50x0.25



UNIT: m

Node	X-Coord	Y-Coord	Z-Coord
4	0.122	3.29961	-0.7
10	0.122	3.29961	5.78501

Additional Info

Beta Angle: 0

Member

Change Beta

Releases:

Start:

End.

Fire Proofing :

Radius of Curvature :

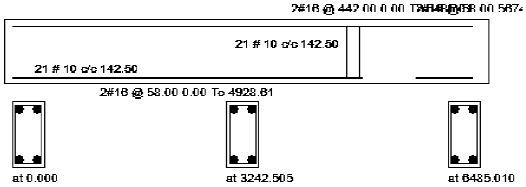
Gamma Angle : deg

Change Releases At Start....

Change Releases At End

Geometry Property Loading Shear Bending Deflection Concrete Design

Beam no. = 11 Design code : ACI-08



Design Load

Mz Kn Met	Dist. Met	Load
7.96	0	27
-0.01	0	4
-18.82	0	28

Design Parameter

Fy(Mpa)	414
Fc(Mpa)	35
Depth(m)	0.5
Width(m)	0.25
Length(m)	6.485009849

Ast.min= **406mm^2**

Hence Provide 3-16 dia-Full Straight Bars Top & Bottom

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8.3.1. Staad Output For B1

```
          BEAM  NO.    11 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

 LEN =   6485. MM  FY =   414.  FC =   35.  MPA, SIZE =   250. X  500. MMS

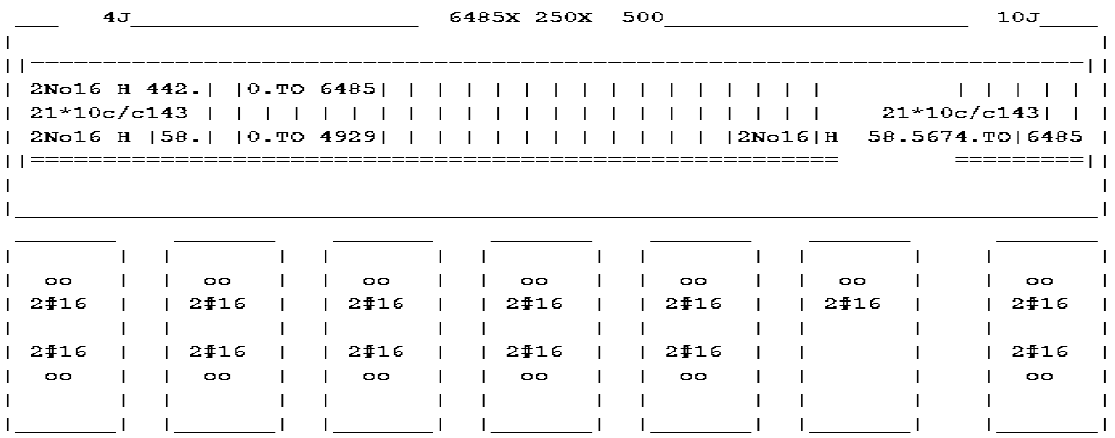
 LEVEL     HEIGHT      BAR INFO        FROM          TO          ANCHOR
                (MM)                      (MM)          (MM)          STA  END
 ____________________________________________________________________________


   1          58.      2 - 16MM          0.          4929.         YES  NO
   2          58.      2 - 16MM       5674.          6485.         NO   YES
   3         442.      2 - 16MM          0.          6485.         YES  YES


              B E A M  N O.    11 D E S I G N  R E S U L T S - SHEAR


 AT START SUPPORT - Vu=     9.70 KNS  Vc=  108.25 KNS  Vs=     0.00 KNS
 Tu=     7.49 KN-MET  Tc=     3.8 KN-MET  Ts=    10.0 KN-MET  LOAD     23
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2803. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    2.47 SQ.CM.

 AT END   SUPPORT - Vu=    12.22 KNS  Vc=  103.14 KNS  Vs=     0.00 KNS
 Tu=     5.34 KN-MET  Tc=     4.0 KN-MET  Ts=     7.1 KN-MET  LOAD     27
 STIRRUPS ARE REQUIRED FOR TORSION.
```


4J
6485X 250X 500
10J
2No16 H 442.
0.TO 6485
21*10c/c143
2No16 H 58.
0.TO 4929
21*10c/c143
2No16 H 58.5674.TO 6485
oo
2#16
2#16
2#16

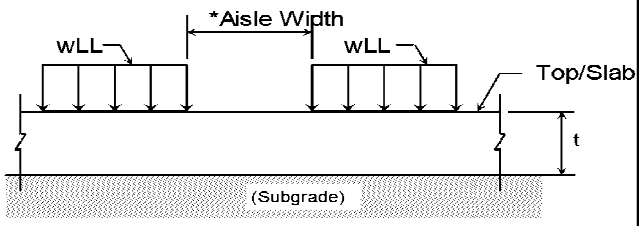
9. CONCRETE GRADE SLAB DESIGN

CONCRETE SLAB ON GRADE ANALYSIS

For Slab Subjected to Stationary Uniformly Distributed Live Loads

Input Data:

Slab Thickness, t =	3.937	in.	100.000	mm		
Concrete Strength, f 'c =	5000	psi	35.000	kpa		
Subgrade Modulus, k =	39.7866	pci	10800	Kn/m3	(AS PER SECTION-7.7-SOIL REPORT)	
Factor of Safety, FS =	1.500		1.500			
Uniform Live Load, wLL =	104.45	psf	5.00	kpa		



Results: **Concrete Slab on Grade with Uniform Loads**

Design Parameters:

Modulus of Rupture, MR =	636.40	psi	4387.79	kpa	MR = 9*SQRT(f 'c)
Low. Bending Stress, Fb =	424.26	psi	2925.19	kpa	Fb = MR/FS
Modulus of Elasticity, Ec =	4030509	psi	27789329	kpa	Ec = 57000*SQRT(f 'c)
Poisson's Ratio, μ =	0.15		0.15		μ = 0.15 (assumed for concrete)
Radius of Stiffness, Lr =	26.94	in.	684.37	mm	Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25
Critical Aisle Width, Wcr =	4.96	ft.	1511.77	mm	Wcr = (2.209*Lr)/12

Stationary Uniformly Distributed Live Loads:

wLL(allow) =	682.05	psf	wLL(allow) = 257.876*Fb*SQRT(k*t/Ec)
	32.65	kpa	**wLL(allow) >= wLL, O.K.**

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 140 mm2/M

Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area

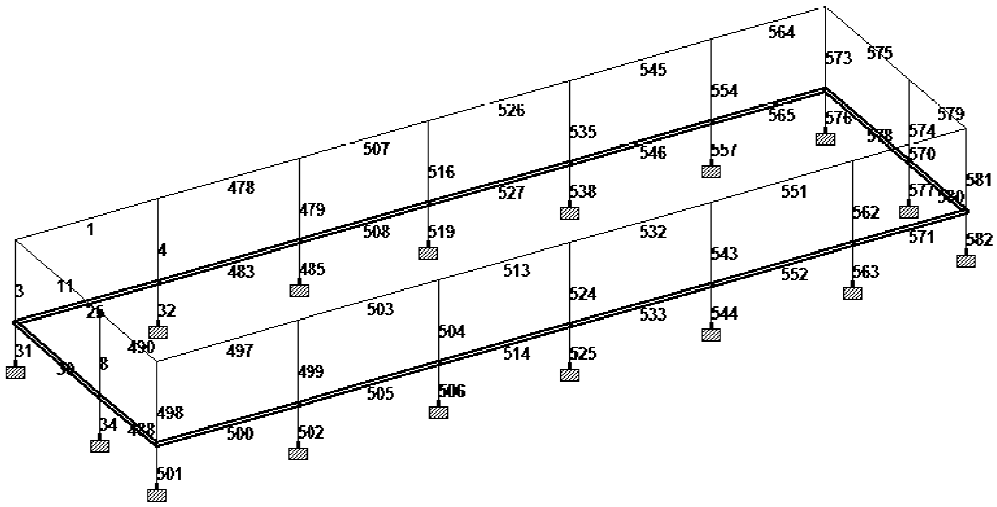
Reinforcement type	f_y, psi	Minimum reinforcement ratio	
Deformed bars	< 60,000	0.0020	
Deformed bars or welded wire reinforcement	≥ 60,000	Greater of:	$\frac{0.0018 \times 60,000}{f_y}$
			0.0014

Provide WWF-8mm Dia@200 c-c Both Ways-TOP LAYER

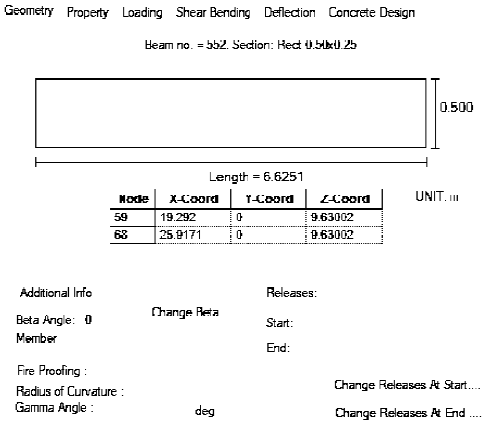
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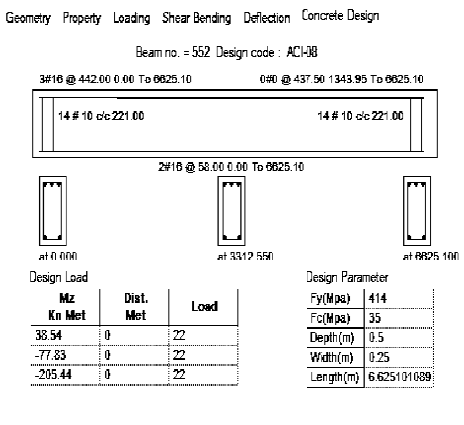
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10. DESIGN OF BEAM-(GB1)



MOMENT ENVELOPE FOR BEAMS GB1





Hence Provide 3-16 dia-Full Straight Bars Top & Bottom

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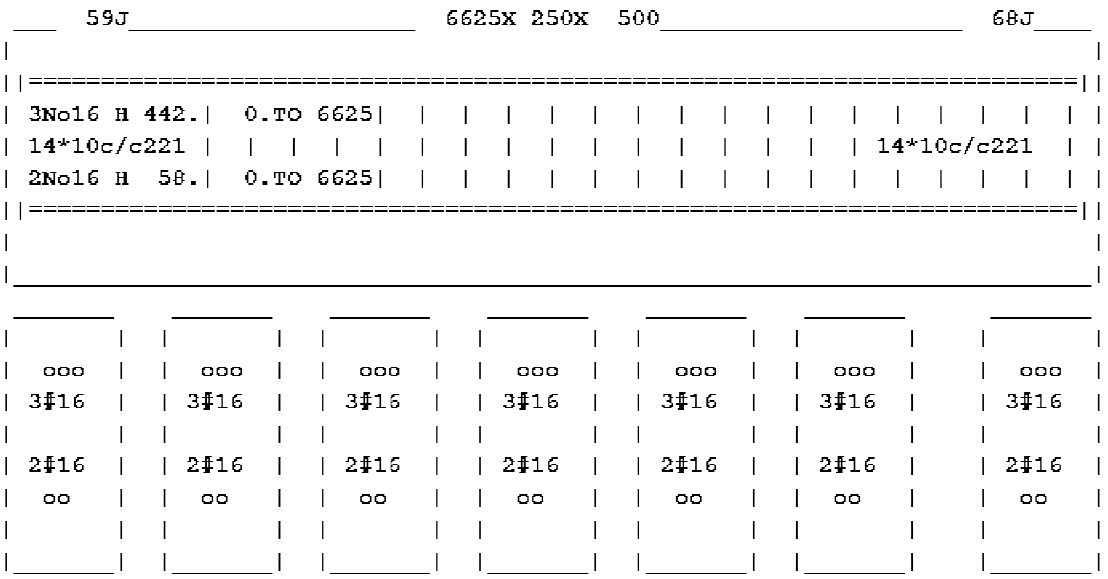
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STAAD OUTPUT FOR GB1

B E A M N O. 552 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   43.28 KNS  Vc=  105.23 KNS  Vs=    0.00 KNS
Tu=    0.22 KN-MET  Tc=    3.7 KN-MET  Ts=    0.0 KN-MET  LOAD     3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 2873. MM

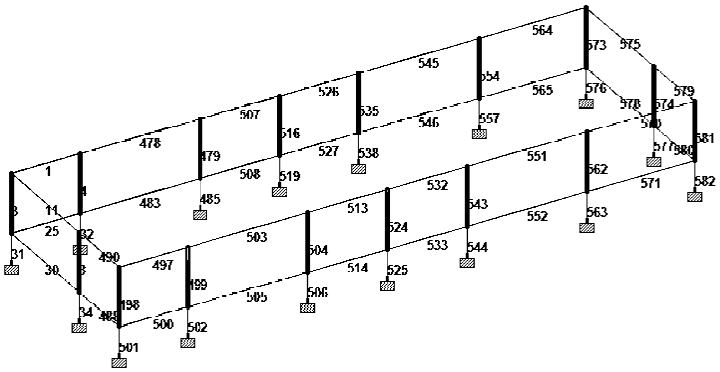
AT END   SUPPORT - Vu=   42.62 KNS  Vc=  105.23 KNS  Vs=    0.00 KNS
Tu=    0.22 KN-MET  Tc=    3.7 KN-MET  Ts=    0.0 KN-MET  LOAD     3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 2873. MM
```



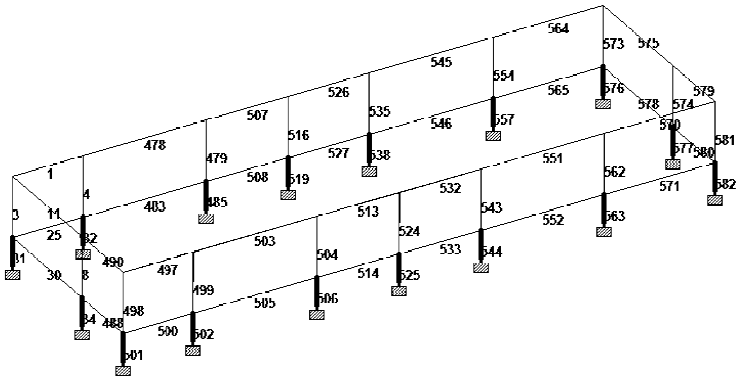
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11. CONCRETE COLUMN DESIGN



Column numbers C1



Column numbers CR

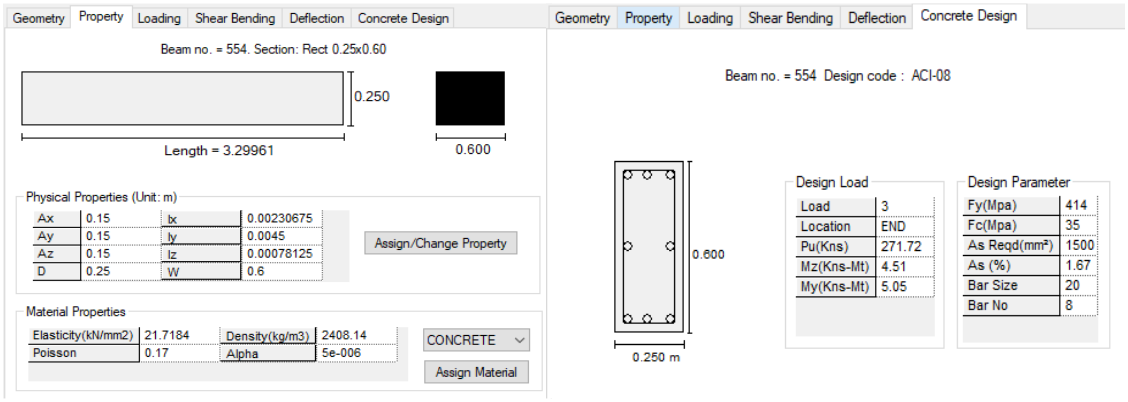
Beam End Summary

All / Summary / Envelope /

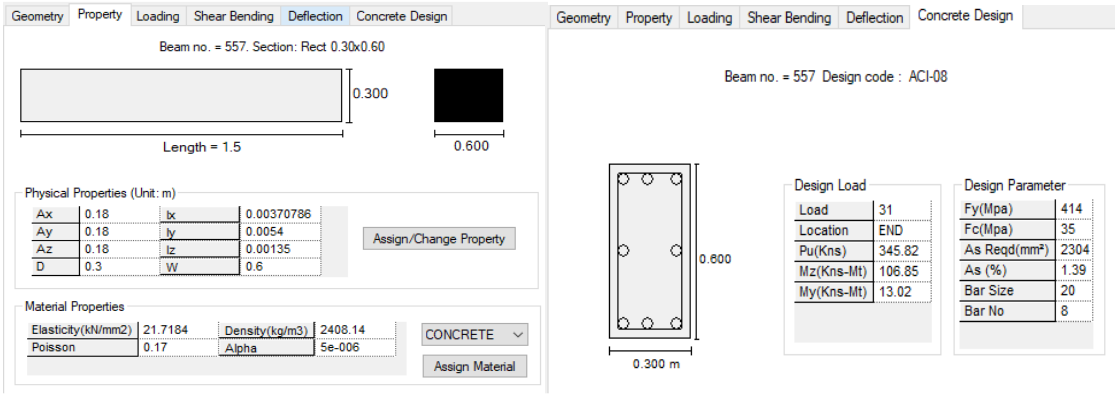
	Beam	L/C	Node	Fx kN	Fy kN	Fz kN	Mx kNm	My kNm	Mz kNm
Max Fx	557	22 SEISMIC	85	524.486	0.878	-11.190	1.288	-5.990	2.212
Min Fx	565	27 GENERAT	62	**-25.642**	51.650	0.071	8.049	-0.115	48.749
Max Fy	503	22 GENERAT	37	46.065	**186.129**	-0.141	-5.682	0.134	194.796
Min Fy	478	22 GENERAT	28	51.218	**-188.683**	0.136	-3.833	0.761	200.297
Max Fz	502	23 GENERAT	35	384.488	0.811	**146.888**	-0.329	-75.935	-3.456
Min Fz	582	23 GENERAT	77	265.433	46.443	**-177.745**	-2.917	78.629	31.113
Max Mx	498	27 GENERAT	34	109.390	-8.537	-14.151	**19.821**	30.665	-10.186
Min Mx	497	27 GENERAT	36	22.203	72.006	-8.727	**-18.155**	15.194	15.071
Max My	502	23 GENERAT	39	392.139	0.811	146.888	-0.329	**144.398**	-4.673
Min My	582	23 GENERAT	78	273.084	46.443	-177.745	-2.917	**-187.989**	-38.551
Max Mz	545	22 GENERAT	61	52.688	-188.318	-0.047	-5.286	0.233	**205.892**
Min Mz	538	31 GENERAT	56	295.980	45.668	-0.454	1.164	-4.472	**-127.130**

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Column Design-C1



Column Design-CR (Column Neck)

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11.1. Staad Output For Column C1

```
     COLUMN  NO.   554  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 600.0 X 250.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  1500.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
------------------------------------------------------------

  8 - 20 MM               1.676        3      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 250.00 MM


=============================================================================


     COLUMN  NO.   557  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 600.0 X 300.0 MMS, TIED
        AREA OF STEEL REQUIRED =  2304.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
------------------------------------------------------------

  8 - 20 MM               1.396       31      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 300.00 MM
```

Hence Provide 8-20 dia-Vertical Reinf for Column Reinforcement-C1 &
Provide 8-25 dia-Vertical Reinf for Column Reinforcement-CR(Column Neck)

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12. CANTILEVER SLAB DESIGN

DESIGN OF CANTILEVER SLAB

Geometrical Properties			Material Properties			
Span to center of support "sp_1"	0.8	met	Weight of reinforced concrete "γ_r"	24		kN/m^3
Clear Span	0.6	met	Strength of concrete "f'_c"	35		N/mm^2
Clear span + depth of slab "sp_2"	0.75	met	Strength of steel "f_y"	420		N/mm^2
Max. eff span le = min (sp_1;sp_2)	0.8	met	Clear cover to concrete("c")	20		mm
Width of Slab "b"	1000	mm	Flexure strength reduction factor	0.9		
Thickness of slab "t"	150	mm	Shear strength reduction factor	0.75		
Eff. Depth "d"	125		Modulus of Elasticity of steel "Es"	2E+05		N/mm^2
Loading			Modulus of Elasticity of concrete	27806		N/mm^2
Self Weight of slab	3.6	kN/m^2	Modular Ratio "n = Es/Ec"	7.19		
Floor Finish	3.5	kN/m^2	**Provided Reinforcement**			
Wall load	0	kN/m^2	Top layer Nos.	6	Φ 10	mm
Total dead Load "U_{dl}"	7.1	kN/m^2	Bottom layer Nos.	0	Φ 12	mm
Live load	1	kN/m^2	Area of Top steel "$A_{s\ t}$"		468	mm^2
Total Ultimate Load "Uu_{tot}"$_{(1.2D+1.6L)}$	10.12	kN/m^2	Area of Bottom steel "$A_{s\ b}$"		0	mm^2
Moment due to Overhang Point Load(0.125*0.15*24*0.6)	0.27	kNm	"db" = 10 mm			
Calculations						
Shear force Vu = U*le	**8.096**	**kN**				
Neg. Moment Mu = U*le^2/2	**3.51**	**kNm**				

Check for Section capacity

Depth of comp. block "a" = $A_{s\ t}$*f_y/(0.85*f'_c*b)	6.61	mm						
Section capacity ΦMn = Φ*$A_{s\ t}$*f_y*(d-a/2)	**21.94**	**kNm**	**O.K**	**>**	**3.51**	**kNm**	(Mu)	

Shear requirements

Shear capacity ΦVc = Φ*0.17*√f'c*b*d	**94.29**	**kN**	**O.K**	**>**	**8.096**	**kN**	(Vu)	

Serviceability requirements

d' = c+d_b/2	26	mm
B = b/(n*$A_{s\ t}$)	0.297	
r = (n-1)*$A_{s\ b}$/(n*$A_{s\ t}$)	0	
k_d = (√(2*d*B(1+r*d'/d)+(1+r)^2)-(1+r))/B	29.01	mm
I_{cr} = bkd^3/3+n*$A_{s\ t}$*(d-k_d)^2+(n-1)*$A_{s\ b}$*(k_d-d').	3.915E+07	mm^4
Calculate Gross moment of inertia		
Area of slab A = t*b	2E+05	mm^2
Depth of neutral axis "y"	75	mm
Distance of top fiber from N.A "y_t"	75	mm
Gross moment of inertia "I_g"	2.813E+08	mm^4
Allowable stress fr = 0.62√f'c	3.67	mm^2
Cracking moment M_{cr} = fr*I_g/y_t	13.75	kNm

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Service moments

Moment due to dead load "M_{dl}"	2.54	kNm
Moment due to live load "Mll"	0.32	kNm
Total Moment Ms = $M_{dl}+M_{ll}$	2.86	kNm

Deflection due to dead + live loads

R(dl+ll) = M_{cr}/Ms	1.000		
$I_e(dl+ll) = R^3*I_g+(1-R^3)*I_{cr}$	2.813E+08	mm^4	
$\delta = K*5/48*Ms*le^2/(Ec*I_e)$	0.08	mm	**K = 3.2 (conservative)**

Deflection due to dead load

R(dl) = M_{cr}/Md	5.411	
$I_e(dl) = R^3*I_g+(1-R^3)*I_{cr}$	3.839E+10	mm^4
$\delta = K*5/48*Md*le^2/(Ec*I_e)$	0.00	mm

Deflection due to live load

Live load deflection = dl+ll deflection - dl deflection 0.08 mm

Allowable immediate deflection due to Live load =
per ACI 318 Tab 9.5b **$I_e/360$ 2.22 mm > 0.08 mm O.K**

Long term deflection

% of liveload sustained "SL"	50	%
$M_{sus} = M_{dl}+SL*M_{ll}$	2.70	kNm
R(sus) = $M_{cr}/Msus$	5.091	
$I_e(sus) = R^3*I_g+(1-R^3)*I_{cr}$	3.198E+10	
$\delta = K*5/48*Msus*le^2/(Ec*I_e)$	0.00	mm

Magnification factor for creep and shrinkage

$\lambda_\Delta = \xi / (1+50*\rho')$ 2.000

Where:-

ξ = 2 for 5 years or more cl.9.5.2.5 ACI 318

$\rho' = A_{s\ b}/(b*d)$ 0

Total deflection = λ_Δ*sus deflection + ll deflection 0.079 mm

Allowable deflection = $I_e/240$ 3.333 mm > 0.079 mm O.K

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13. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 350 Kpa. (Refer Doc-MD-216-3000-EG-CV-CAL-1008-A01 for Details)

[redacted]

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^2)
Village Area	BH-21	Village	Isolated Square	1.20	1.0 x 1.0	1000	350	350	25	42,000	32,200
				1.50		1100	500	500	25	60,000	38,900
				1.20	2.0 x 2.0	700	580	580	25	69,600	37,500
				1.50		850	750	750	25	90,000	44,200
				1.20	3.0 x 3.0	600	750	600	25	72,000	42,800
				1.50		700	930	700	25	84,000	49,500
				1.20	4.0 x 4.0	550	910	550	25	66,000	48,100
				1.50		625	1090	625	25	75,000	54,800
				1.20	5.0 x 5.0	475	1060	475	25	57,000	53,400
				1.50		550	1240	550	25	66,000	60,200

Refer Attachment- II for Foundation Design calculations

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REFERENCES:

BUILDING PLANS	MD-216-6000-EG-AR-GAD-3087
OVERALL LAYOUT PLAN	MD-216-6000-EG-GAD-1535
LAYOUT & DETAILS OF FOUNDATION, COLUMN & GRADE BEAM	MD-216-6000-EG-CV-DED-1484
LAYOUT & DETAILS OF ROOF & GRADE SLAB	MD-216-6000-EG-CV-DED-1485
GEOTECHNICAL REPORT	MD-216-3000-EG-CV-RPT-1008
STRUCTURAL DESIGN CRITERIA	MD-216-0000-EG-ST-CRT-0001

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ATTACHMENT 1 STAAD INPUT DATA

STAAD SPACE

START JOB INFORMATION

ENGINEER DATE 18-Oct-19

END JOB INFORMATION

INPUT WIDTH 79

UNIT METER KN

JOINT COORDINATES

1 0.122 0 -0.7; 2 3.92201 0 -0.7; 4 0.122 3.29961 -0.7; 5 3.92201 3.29961 -0.7;
7 0.122 0 5.78501; 10 0.122 3.29961 5.78501; 23 0.122 -1.5 -0.7;
24 3.92201 -1.5 -0.7; 26 0.122 -1.5 5.78501; 28 10.547 3.29961 -0.7;
29 10.547 0 -0.7; 32 10.547 -1.5 -0.7; 34 0.122 0 9.63002;
35 3.92201 0 9.63002; 36 0.122 3.29961 9.63002; 37 3.92201 3.29961 9.63002;
38 0.122 -1.5 9.63002; 39 3.92201 -1.5 9.63002; 40 10.547 3.29961 9.63002;
41 10.547 0 9.63002; 42 10.547 -1.5 9.63002; 43 14.947 3.29961 -0.7;
44 14.947 0 -0.7; 47 14.947 -1.5 -0.7; 49 14.947 3.29961 9.63002;
50 14.947 0 9.63002; 51 14.947 -1.5 9.63002; 52 19.292 3.29961 -0.7;
53 19.292 0 -0.7; 56 19.292 -1.5 -0.7; 58 19.292 3.29961 9.63002;
59 19.292 0 9.63002; 60 19.292 -1.5 9.63002; 61 25.9171 3.29961 -0.7;
62 25.9171 0 -0.7; 65 25.9171 -1.5 -0.7; 67 25.9171 3.29961 9.63002;
68 25.9171 0 9.63002; 69 25.9171 -1.5 9.63002; 70 31.7971 3.29961 -0.7;
71 31.7971 0 -0.7; 72 31.7971 3.29961 5.78501; 73 31.7971 0 5.78501;
74 31.7971 -1.5 -0.7; 75 31.7971 -1.5 5.78501; 76 31.7971 3.29961 9.63002;
77 31.7971 0 9.63002; 78 31.7971 -1.5 9.63002;

MEMBER INCIDENCES

1 4 5; 3 1 4; 4 2 5; 8 7 10; 11 4 10; 25 1 2; 30 7 1; 31 1 23; 32 2 24;
34 7 26; 478 5 28; 479 29 28; 483 2 29; 485 29 32; 488 7 34; 490 10 36;
497 36 37; 498 34 36; 499 35 37; 500 35 34; 501 34 38; 502 35 39; 503 37 40;
504 41 40; 505 41 35; 506 41 42; 507 28 43; 508 29 44; 513 40 49; 514 41 50;
516 44 43; 519 44 47; 524 50 49; 525 50 51; 526 43 52; 527 44 53; 532 49 58;

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533 50 59; 535 53 52; 538 53 56; 543 59 58; 544 59 60; 545 52 61; 546 53 62;

551 58 67; 552 59 68; 554 62 61; 557 62 65; 562 68 67; 563 68 69; 564 61 70;

565 62 71; 570 67 76; 571 68 77; 573 71 70; 574 73 72; 575 70 72; 576 71 74;

577 73 75; 578 71 73; 579 72 76; 580 73 77; 581 77 76; 582 77 78;

DEFINE MATERIAL START

ISOTROPIC CONCRETE

E 2.17184e+007

POISSON 0.17

DENSITY 23.6158

ALPHA 5e-006

DAMP 0.05

TYPE CONCRETE

STRENGTH FCU 27578.9

END DEFINE MATERIAL

MEMBER PROPERTY AMERICAN

31 32 34 485 501 502 506 519 525 538 544 557 563 576 577 582 PRIS YD 0.3 ZD 0.6

1 478 497 503 507 513 526 532 545 551 564 570 PRIS YD 0.5 ZD 0.25

MEMBER PROPERTY AMERICAN

25 30 483 488 500 505 508 514 527 533 546 552 565 571 578 -

580 PRIS YD 0.5 ZD 0.25

MEMBER PROPERTY AMERICAN

11 490 575 579 PRIS YD 0.5 ZD 0.25

MEMBER PROPERTY AMERICAN

3 4 8 479 498 499 504 516 524 535 543 554 562 573 574 581 PRIS YD 0.25 ZD 0.6

CONSTANTS

BETA 90 MEMB 4 32 479 485 499 502 504 506 516 519 524 525 535 538 543 544 -

554 557 562 563

MATERIAL CONCRETE ALL

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SUPPORTS

23 24 26 32 38 39 42 47 51 56 60 65 69 74 75 78 FIXED

****

MEMBER CRACKED

3 4 8 31 32 34 REDUCTION RIY 0.700000 RIZ 0.700000

1 11 25 30 REDUCTION RIY 0.350000 RIZ 0.350000

MEMBER OFFSET

478 503 507 513 526 532 545 551 564 570 START 0.3 0 0

1 478 497 503 507 513 526 532 545 551 END -0.3 0 0

11 490 575 579 START 0 0 0.3

11 490 575 579 END 0 0 -0.3

1 497 START 0.125 0 0

564 570 END -0.125 0 0

DEFINE IBC 2003

SDS 0.1 SD1 0.03 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2

SELFWEIGHT 1

MEMBER WEIGHT

25 30 483 488 500 505 508 514 527 533 546 552 565 571 578 580 UNI 12

1 478 497 503 507 513 526 532 545 551 564 570 UNI 37.2

DEFINE WIND LOAD

TYPE 1 WIND X WINDWARD

<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!

ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -

1 3.500 M 33.000 M 10.000 M 2.000 0.010 0 -

0 0 1 1 1.030 1.000 1.000 0.850 0 -

1 0 1 0.850 0.800 -0.550

!> END GENERATED DATA BLOCK

INT 1.59699 1.59699 1.31588 HEIG 0 4.572 4.572

EXP 0.8 JOINT 1 2 4 5 7 10 23 24 26 28 29 32 38 39 42 47 51 56 60 65 69 74 -

75 78

TYPE 2 WIND X LEEWARD

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<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.500 M 33.000 M 10.000 M 2.000 0.010 1 -
0 0 1 1 1.030 1.000 1.000 0.850 0 -
1 0 1 0.850 -0.200 0.550
!> END GENERATED DATA BLOCK
INT -0.934826 -0.934826 -0.934826 HEIG 0 3.5 3.5
EXP 0.8 JOINT 1 2 4 5 7 10 23 24 26 28 29 32 38 39 42 47 51 56 60 65 69 74 -
75 78
TYPE 3 WIND Z WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 6.000 M 10.000 M 2.000 0.010 0 -
0 0 1 1 1.030 1.000 1.000 0.850 0 -
1 0 1 0.850 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.59699 1.59699 1.59699 HEIG 0 4.572 4.572
EXP 0.8 JOINT 1 2 4 5 7 10 23 24 26 28 29 32 38 39 42 47 51 56 60 65 69 74 -
75 78
TYPE 4 WIND Z LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2010:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 6.000 M 10.000 M 2.000 0.010 1 -
0 0 1 1 1.030 1.000 1.000 0.850 0 -
1 0 1 0.850 -0.500 0.550
!> END GENERATED DATA BLOCK
INT -1.26591 -1.26591 -1.26591 HEIG 0 4 4
EXP 0.8 JOINT 1 2 4 5 7 10 23 24 26 28 29 32 38 39 42 47 51 56 60 65 69 74 -
75 78
LOAD 1 LOADTYPE Seismic TITLE EQK LOAD Z
IBC LOAD Z 1

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PERFORM ANALYSIS

CHANGE

LOAD 2 LOADTYPE Seismic TITLE EQK LOAD X

IBC LOAD X 1

PERFORM ANALYSIS

CHANGE

LOAD 3 LOADTYPE Dead TITLE DL

SELFWEIGHT Y -1

MEMBER LOAD

25 30 483 488 500 505 508 514 527 533 546 552 565 571 578 580 UNI GY -12

1 478 497 503 507 513 526 532 545 551 564 570 UNI GY -37.2

11 490 UMOM GZ 1

575 579 UMOM GZ -1

1 478 507 526 545 564 UMOM GX -1

497 503 513 532 551 570 UMOM GX 1

LOAD 4 LOADTYPE Live TITLE LL

MEMBER LOAD

1 478 497 503 507 513 526 532 545 551 564 570 UNI GY -2.5

11 490 UMOM GZ 0.1

575 579 UMOM GZ -0.1

1 478 507 526 545 564 UMOM GX -0.1

497 503 513 532 551 570 UMOM GX 0.1

LOAD 5 LOADTYPE Wind TITLE WL+X

WIND LOAD X 1 TYPE 1 XR 0 0.2 YR 0 4 ZR -0.7 10

WIND LOAD -X 1 TYPE 2 XR 31.7 33 YR 0 4

WIND LOAD Z 1 TYPE 3 XR 0 33 YR 0 4 ZR -0.7 -0.6

WIND LOAD -Z 1 TYPE 4 YR 0 4 ZR 9.63 10

LOAD 6 LOADTYPE Wind TITLE WL+Z

WIND LOAD Z 1 TYPE 3 XR 0 33 YR 0 4 ZR -0.7 -0.6

WIND LOAD -Z 1 TYPE 4 XR 0 32 YR 0 4 ZR 9.63 10

WIND LOAD X 1 TYPE 1 XR 0.1 0.122 YR 0 4 ZR -0.7 10

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WIND LOAD -X 1 TYPE 2 XR 31 33 YR 0 4 ZR -0.7 10

LOAD 7 LOADTYPE Temperature TITLE TEMP

TEMPERATURE LOAD

1 3 4 8 11 25 30 TO 32 34 478 479 483 485 488 490 497 TO 508 513 514 516 519 -

524 TO 527 532 533 535 538 543 TO 546 551 552 554 557 562 TO 565 570 571 -

573 TO 582 TEMP 25

LOAD COMB 8 SLS: (DL)

3 1.0

LOAD COMB 9 SLS: (DL+T)

3 1.0 7 1.0

LOAD COMB 10 SLS: (D+LR)

3 1.0 4 1.0

LOAD COMB 11 SLS: (D+0.75LR)

3 1.0 4 0.75

LOAD COMB 12 SLS: (D+0.7EZ)

3 1.0 1 0.7

LOAD COMB 13 SLS: (D+0.7EX)

3 1.0 2 0.7

LOAD COMB 14 SLS: (D+0.75WX+0.75LR)

3 1.0 5 0.75 4 0.75

LOAD COMB 15 SLS: (D+0.75WZ+0.75LR)

3 1.0 6 0.75 4 0.75

LOAD COMB 16 SLS: (0.6D+WX)

3 0.6 5 1.0

LOAD COMB 17 SLS: (0.6D+WZ)

3 0.6 6 1.0

LOAD COMB 18 SLS: (0.6D+0.7EX)

3 0.6 2 0.7

LOAD COMB 19 SLS: (0.6D+0.7EZ)

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3 0.6 1 0.7

LOAD COMB 20 SLS: (D+WX)

3 1.0 5 1.0

LOAD COMB 21 SLS: (D+WZ)

3 1.0 6 1.0

LOAD COMB 22 ULS: (1.4 DL)

3 1.4

LOAD COMB 23 ULS: (1.4D+1.4T+0.5LR)

3 1.4 4 0.5 7 1.4

LOAD COMB 24 ULS: (1.4D+1.4T+0.5LR)

3 1.4 4 0.5 7 1.4

LOAD COMB 25 ULS: (1.2D+0.80WX+1.6LR)

3 1.2 5 0.8 4 1.6

LOAD COMB 26 ULS: (1.2D+1.6WZ+0.5LR)

3 1.2 6 1.6 4 0.5

LOAD COMB 27 ULS: (1.2D+1.6WX+0.5LR)

3 1.2 4 0.5 5 1.6

LOAD COMB 28 ULS: (1.2D+1.6WZ+0.5LR)

3 1.2 4 0.5 6 1.6

LOAD COMB 29 ULS: (1.2D+EX)

3 1.2 2 1.0

LOAD COMB 30 ULS: (1.2D+EZ)

3 1.2 1 1.0

LOAD COMB 31 ULS: (0.90D+1.6WX)

3 0.9 5 1.6

LOAD COMB 32 ULS: (0.90D+1.6WZ)

3 0.9 6 1.6

LOAD COMB 33 ULS: (0.90D+EZ)

3 0.9 1 1.0

LOAD COMB 34 ULS: (0.90D+EX)

3 0.9 2 1.0

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PERFORM ANALYSIS

START CONCRETE DESIGN

CODE ACI

FC 35000 ALL

CLB 0.04 ALL

CLS 0.04 ALL

CLT 0.04 ALL

MAXMAIN 20 MEMB 3 4 8 31 32 34 479 485 498 499 501 502 504 506 516 519 524 -

525 535 538 543 544 554 557 562 563 573 574 576 577 581 582

MINMAIN 20 MEMB 3 4 8 31 32 34 479 485 498 499 501 502 504 506 516 519 524 -

525 535 538 543 544 554 557 562 563 573 574 576 577 581 582

MAXMAIN 25 MEMB 1 11 478 490 497 503 507 513 526 532 545 551 564 570 575 579

MINMAIN 20 MEMB 1 11 478 490 497 503 507 513 526 532 545 551 564 570 575 579

MAXMAIN 16 MEMB 11 490 575 579

MINMAIN 16 MEMB 11 490 575 579

MAXMAIN 20 MEMB 25 30 483 488 500 505 508 514 527 533 546 552 565 571 578 580

MINMAIN 16 MEMB 25 30 483 488 500 505 508 514 527 533 546 552 565 571 578 580

MINSEC 10 ALL

EFACE 0.74 MEMB 1 478 497 503 507 513 526 532 545 551

SFACE 0.74 MEMB 478 503 507 513 526 532 545 551 564 570

EFACE 0.74 MEMB 11 490 575 579

SFACE 0.74 MEMB 11 490 575 579

EFACE 0.69 MEMB 564 570

SFACE 0.69 MEMB 1 497

DESIGN BEAM 1 11 25 30 478 483 488 490 497 500 503 505 507 508 513 514 526 -

527 532 533 545 546 551 552 564 565 570 571 575 578 TO 580

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DESIGN COLUMN 3 4 8 31 32 34 479 485 498 499 501 502 504 506 516 519 524 525 -

535 538 543 544 554 557 562 563 573 574 576 577 581 582

END CONCRETE DESIGN

FINISH

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ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS

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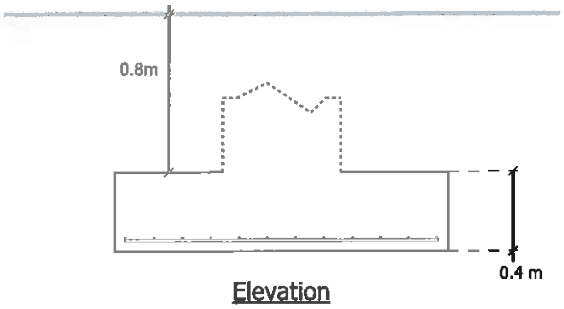
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Isolated Footing Design(ACI 318-05)

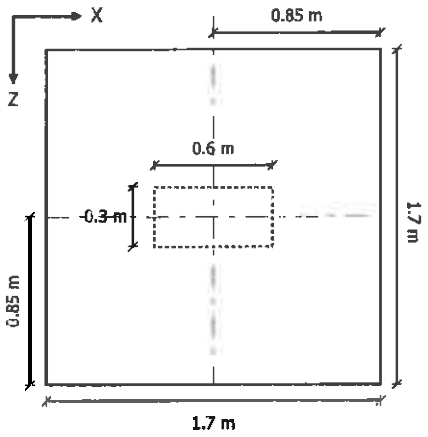
Footing No.	Group ID	Foundation Geometry		
		Length	Width	Thickness
51	1	1.700m	1.700m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
51	12 - ϕ12	12 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F1-GRID-3 & D



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1700.000mm
Footing Width - Z (Fw) : 1700.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.600m

Column Width - Z (B_{col}) : 0.300m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 50.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.580
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.700m

Initial Width (W_o) = 1.700m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
101	SL:1.000 x DL	1.00	1.00	1.00
102	SL:1.000 x DL+1.000 x LL	1.00	1.00	1.00
105	SL:1.000 x DL+0.750 x LL	1.00	1.00	1.00
107	SL:1.000 x DL+1.000 x WLX	1.00	1.00	1.00
109	SL:1.000 x DL+1.000 x WLZ	1.00	1.00	1.00
111	SL:1.000 x DL+0.700 x ELX	1.00	1.00	1.00
113	SL:1.000 x DL+0.700 x ELZ	1.00	1.00	1.00
115	SL:1.000 x DL+0.750 x LL+0.750 x WLX	1.00	1.00	1.00
119	SL:1.000 x DL+0.750 x LL+0.750 x WLZ	1.00	1.00	1.00
123	SL:1.000 x DL+0.750 x LL+0.525 x ELX	1.00	1.00	1.00
127	SL:1.000 x DL+0.750 x LL+0.525 x ELZ	1.00	1.00	1.00
131	SL:0.600 x DL+1.000 x WLX	1.00	1.00	0.60
133	SL:0.600 x DL+1.000 x WLZ	1.00	1.00	0.60
135	SL:0.600 x DL+0.700 x ELX	1.00	1.00	0.60
137	SL:0.600 x DL+0.700 x ELZ	1.00	1.00	0.60

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
201	UL:1.400 x DL	1.00	1.00	1.40
202	UL:1.200 x DL+1.600 x LL	1.00	1.00	1.20
204	UL:1.200 x DL+1.000 x LL	1.00	1.00	1.20
206	UL:1.200 x DL+0.800 x WLX	1.00	1.00	1.20
208	UL:1.200 x DL+0.800 x WLZ	1.00	1.00	1.20
214	UL:1.200 x DL+1.000 x LL+1.600 x WLX	1.00	1.00	1.20
218	UL:1.200 x DL+1.000 x LL+1.600 x WLZ	1.00	1.00	1.20
222	UL:1.200 x DL+1.000 x LL+1.000 x ELX	1.00	1.00	1.20
224	UL:1.200 x DL+1.000 x LL+1.000 x ELZ	1.00	1.00	1.20
226	UL:0.900 x DL+1.600 x WLX	1.00	1.00	0.90
228	UL:0.900 x DL+1.600 x WLZ	1.00	1.00	0.90
230	UL:0.900 x DL+1.000 x ELX	1.00	1.00	0.90
232	UL:0.900 x DL+1.000 x ELZ	1.00	1.00	0.90

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	248.894	-1.115	0.091	4.701	2.210
102	259.142	-1.197	0.101	5.169	2.341
105	256.580	-1.177	0.098	5.052	2.308
107	248.872	2.779	19.660	66.558	-3.362
109	248.872	2.779	19.660	66.558	-3.362
111	248.765	7.100	0.091	4.714	-10.201
113	248.883	-1.101	6.570	32.598	2.189
115	256.564	1.744	14.775	51.445	-1.871
119	256.564	1.744	14.775	51.445	-1.871
123	256.483	4.984	0.098	5.062	-7.000
127	256.571	-1.167	4.957	25.975	2.293
131	149.315	3.225	19.623	64.678	-4.246
133	149.315	3.225	19.623	64.678	-4.246
135	149.207	7.546	0.055	2.833	-11.085
137	149.325	-0.655	6.534	30.717	1.305

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	348.452	-1.561	0.128	6.581	3.094

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
202	315.069	-1.470	0.124	6.391	2.862
204	308.921	-1.420	0.119	6.109	2.783
206	298.656	1.778	15.765	55.127	-1.806
208	298.656	1.778	15.765	55.127	-1.806
214	308.886	4.811	31.429	105.082	-6.133
218	308.886	4.811	31.429	105.082	-6.133
222	308.736	10.315	0.119	6.128	-14.947
224	308.904	-1.401	9.374	45.962	2.754
226	223.970	5.228	31.392	103.203	-6.927
228	223.970	5.228	31.392	103.203	-6.927
230	223.820	10.732	0.082	4.249	-15.741
232	223.988	-0.984	9.338	44.084	1.960

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 2.890m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.599m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.700 m Governing Load Case : # 101

Width (W_2) = 1.700 m Governing Load Case : # 101

Depth (D_2) = 0.400 m Governing Load Case : # 214

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 2.890 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 27.743 kN

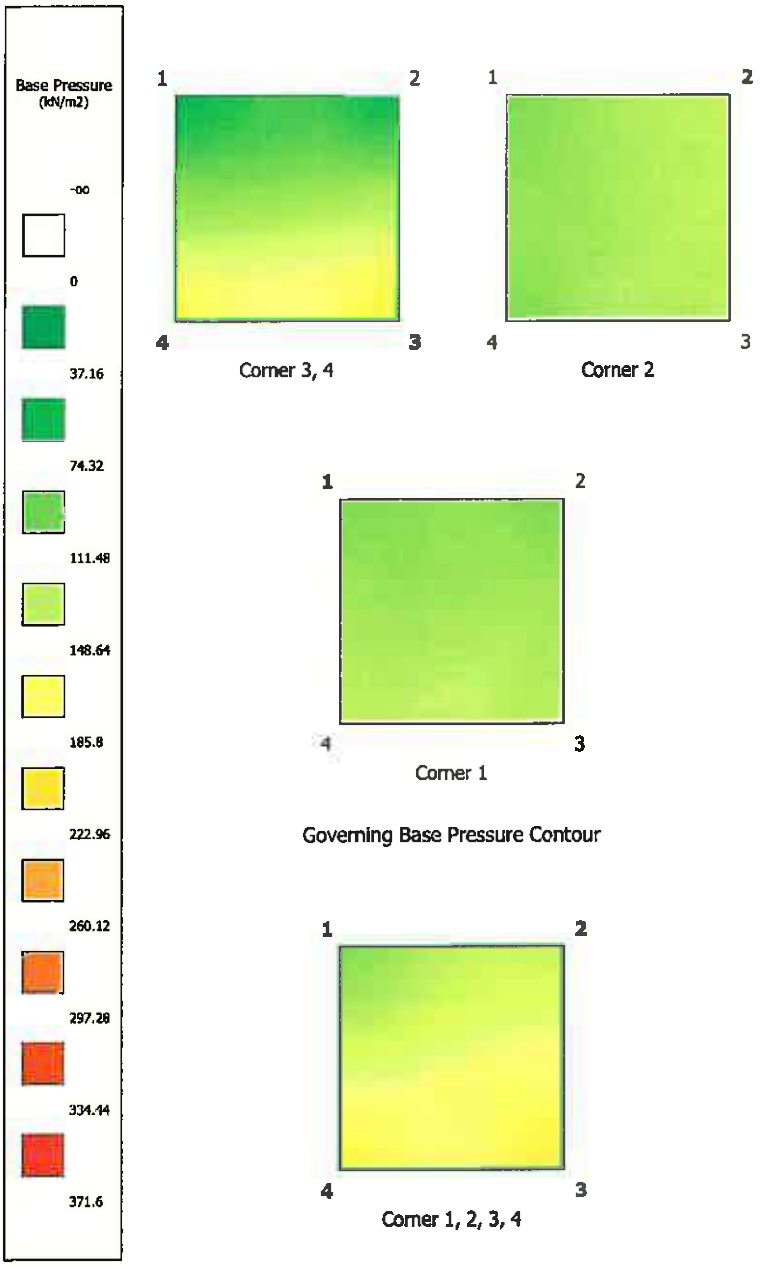
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 39.023 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



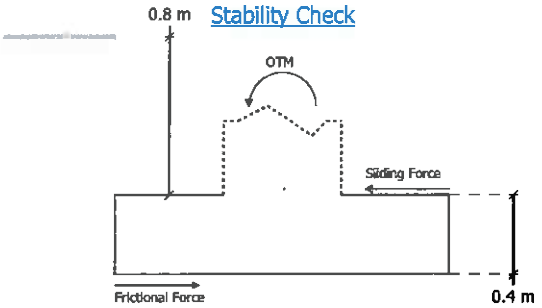
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
102	**109.8525**	102.9646	115.6891	122.5771	0.000
111	87.4522	**119.3056**	130.9080	99.0546	0.000
107	12.8633	23.7916	**205.5712**	194.6430	0.000
107	12.8633	23.7916	205.5712	**194.6430**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
102	**109.8525**	102.9646	115.6891	122.5771
111	87.4522	**119.3056**	130.9080	99.0546
107	12.8633	23.7916	**205.5712**	194.6430
107	12.8633	23.7916	205.5712	**194.6430**

Stability Check



	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
101	164.205	2002.312	231.444	1.500	56.985	101.022	1.500
102	157.865	1878.859	222.470	1.500	53.492	98.234	1.500
105	159.366	1907.563	224.596	1.500	54.304	98.900	1.500
107	65.867	9.312	13.039	1.500	3.627	59.964	1.500
109	65.867	9.312	13.039	1.500	3.627	59.964	1.500
111	25.776	2003.065	36.449	1.500	56.809	20.565	1.500
113	166.211	27.864	38.863	1.500	7.663	102.014	1.500
115	107.530	12.693	17.826	1.500	4.820	106.995	1.500
119	107.530	12.693	17.826	1.500	4.820	106.995	1.500
123	37.614	1908.036	53.184	1.500	54.186	30.550	1.500
127	160.744	37.829	52.075	1.500	9.889	99.594	1.500
131	36.049	5.925	8.269	1.500	2.363	30.777	1.500
133	36.049	5.925	8.269	1.500	2.363	30.777	1.500
135	15.400	2121.041	21.779	1.500	59.995	12.075	1.500
137	177.394	17.797	25.043	1.500	5.142	108.702	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 135
Governing Disturbing Force : 7.546kN
Governing Restoring Force : 116.211kN
Minimum Sliding Ratio for the Critical Load Case : 15.400
Critical Load Case for Overturning about X-Direction : 131
Governing Overturning Moment : 72.527kNm
Governing Resisting Moment : 171.389kNm
Minimum Overturning Ratio for the Critical Load Case : 2.363

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 131
Governing Disturbing Force : 19.623kN
Governing Restoring Force : 116.273kN
Minimum Sliding Ratio for the Critical Load Case : 5.925
Critical Load Case for Overturning about Z-Direction : 135
Governing Overturning Moment : -14.103kNm
Governing Resisting Moment : 170.306kNm
Minimum Overturning Ratio for the Critical Load Case : 12.075

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

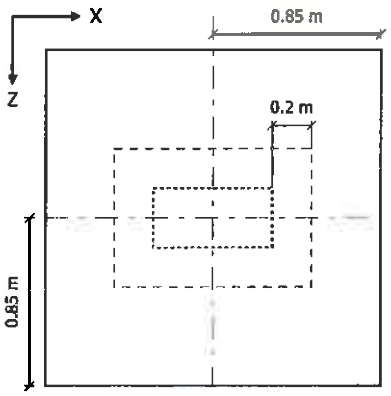
Critical Load Case for Sliding along Resultant Direction : 131
Governing Disturbing Force : 19.887kN
Governing Restoring Force : 164.435kN
Minimum Sliding Ratio for the Critical Load Case : 8.269

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimension Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 348.962kN, Load Case # 201

$2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.176m

$0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2146.802kN

$0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 3398.656kN

$0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2146.802kN

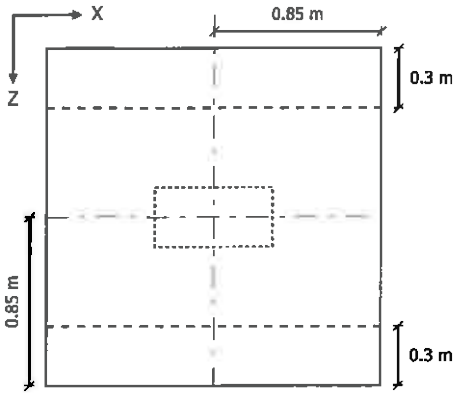
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1610.102kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

$2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 574.554kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.356m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

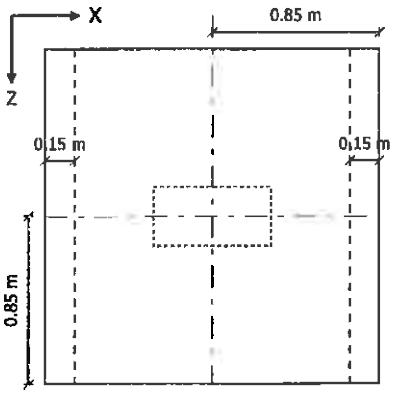
From above calculations, 0.75 X V_c = 430.916 kN

Critical load case for V_{ux} is # 214 $V_{ux} = V_{ux}|_{x=D_x}$ = 149.666 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 574.554 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{zd}$ = 0.206 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

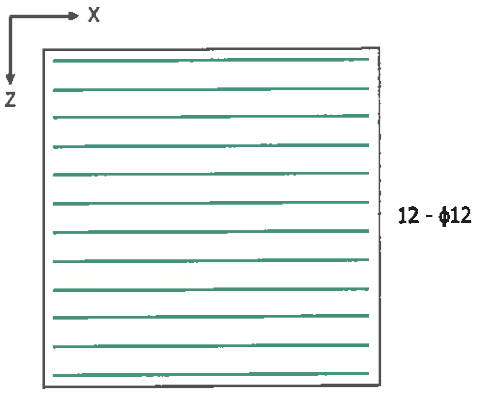
From above calculations, 0.75 X V_c = 430.916 kN

Critical load case for V_{uz} is # 201 $V_{uz} = V_{uz}|_{z=D_z}$ = 54.949 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 201

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f'_c \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00166
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.550 m
Ultimate moment,	$M_u\big\|_{z=D_z}$ =	40.233 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\phi}$ =	44.704 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00053
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00046
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	ρ x W x d_{eff} =	1131.328 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 50.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) - C_{cover}$ =	0.500	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	12	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1357.159	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.344	m
Reinforcement ratio, P =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00232	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

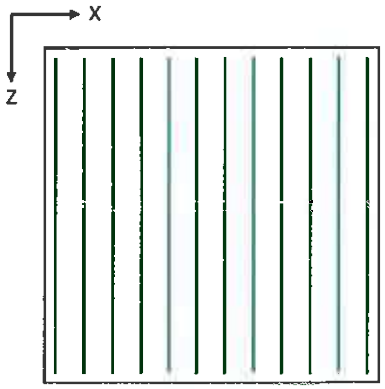
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 50.000mm

Provided Steel Area / Required Steel Area = 1.200

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



12 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 214

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.332	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f'_c \times \left(\frac{87}{87 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl.7.12.2, ρ_{min} =		0.00172	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{(0.85 \times F'_c)}$ =	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ =	1.000	m
Ultimate moment,	$M_u\big\vert_{x=D_x}$ =	100.056	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	111.174	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00143	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00118	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	ρ x W x d_{eff} =	1170.668	mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 50.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) - C_{cover}$ =	0.650	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	12	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

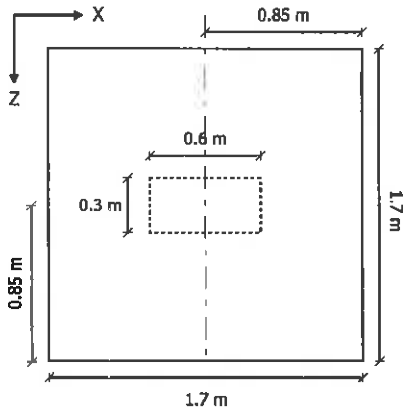
Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1357.159	mm2
d_{eff} =	D - C_{cover} - 1.5 X (dia. of one bar) =	0.332	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00240	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

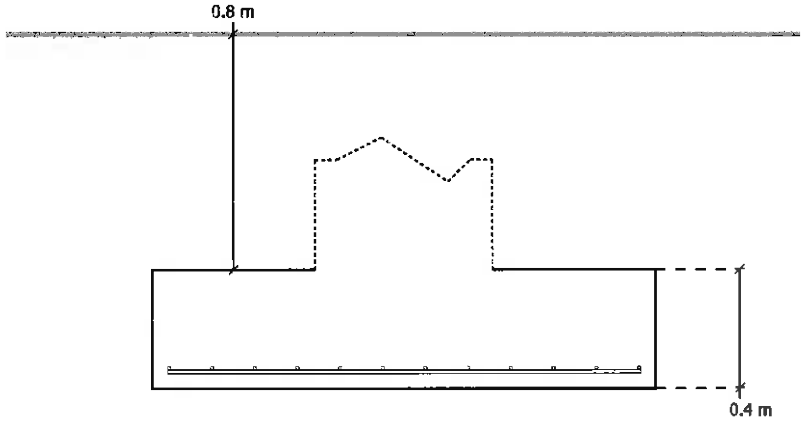
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 50.000mm

Provided Steel Area / Required Steel Area = 1.159

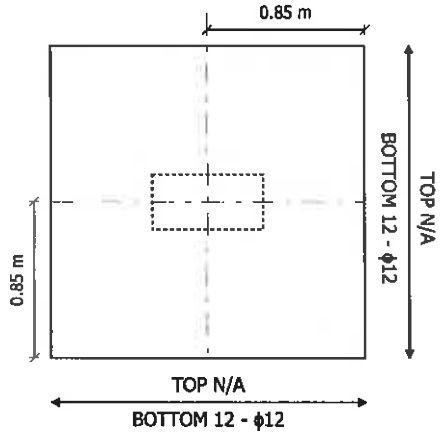
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.7 m	1.7 m	0.4 m	N/A	N/A	12 - ϕ12	12 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

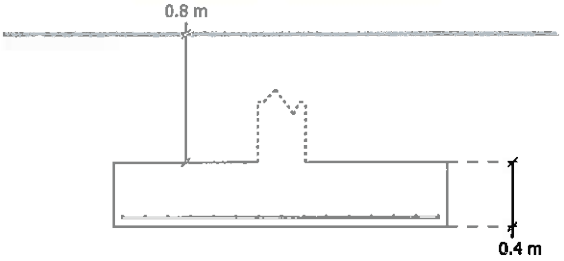
Print Calculation Sheet

Isolated Footing Design(ACI 318-08)

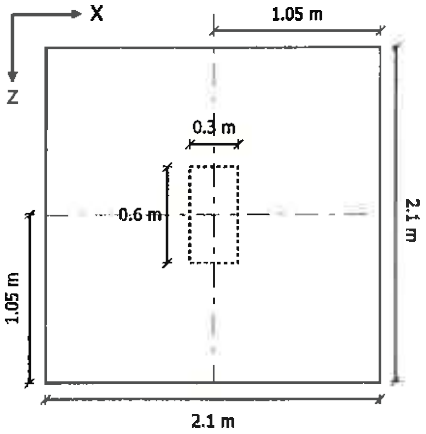
Footing No.	Group ID	Foundation Geometry		
		Length	Width	Thickness
26	1	2.100m	2.100m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
26	15 - ϕ12	15 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F2-GRID-2 & A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 2100.000mm
Footing Width - Z (Fw) : 2100.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{cx}) : 0.300m

Column Width - Z (B_{col}) : 0.600m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 50.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.580
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 2.100m

Initial Width (W_o) = 2.100m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
101	SL:1.000 x DL	1.00	1.00	1.00
102	SL:1.000 x DL+1.000 x LL	1.00	1.00	1.00
105	SL:1.000 x DL+0.750 x LL	1.00	1.00	1.00
107	SL:1.000 x DL+1.000 x WLX	1.00	1.00	1.00
109	SL:1.000 x DL+1.000 x WLZ	1.00	1.00	1.00
111	SL:1.000 x DL+0.700 x ELX	1.00	1.00	1.00
113	SL:1.000 x DL+0.700 x ELZ	1.00	1.00	1.00
115	SL:1.000 x DL+0.750 x LL+0.750 x WLX	1.00	1.00	1.00
119	SL:1.000 x DL+0.750 x LL+0.750 x WLZ	1.00	1.00	1.00
123	SL:1.000 x DL+0.750 x LL+0.525 x ELX	1.00	1.00	1.00
127	SL:1.000 x DL+0.750 x LL+0.525 x ELZ	1.00	1.00	1.00
131	SL:0.600 x DL+1.000 x WLX	1.00	1.00	0.60
133	SL:0.600 x DL+1.000 x WLZ	1.00	1.00	0.60
135	SL:0.600 x DL+0.700 x ELX	1.00	1.00	0.60
137	SL:0.600 x DL+0.700 x ELZ	1.00	1.00	0.60

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
201	UL:1.400 x DL	1.00	1.00	1.40
202	UL:1.200 x DL+1.600 x LL	1.00	1.00	1.20
204	UL:1.200 x DL+1.000 x LL	1.00	1.00	1.20
206	UL:1.200 x DL+0.800 x WLX	1.00	1.00	1.20
208	UL:1.200 x DL+0.800 x WLZ	1.00	1.00	1.20
214	UL:1.200 x DL+1.000 x LL+1.600 x WLX	1.00	1.00	1.20
218	UL:1.200 x DL+1.000 x LL+1.600 x WLZ	1.00	1.00	1.20
222	UL:1.200 x DL+1.000 x LL+1.000 x ELX	1.00	1.00	1.20
224	UL:1.200 x DL+1.000 x LL+1.000 x ELZ	1.00	1.00	1.20
226	UL:0.900 x DL+1.600 x WLX	1.00	1.00	0.90
228	UL:0.900 x DL+1.600 x WLZ	1.00	1.00	0.90
230	UL:0.900 x DL+1.000 x ELX	1.00	1.00	0.90
232	UL:0.900 x DL+1.000 x ELZ	1.00	1.00	0.90

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	115.552	0.299	17.173	8.268	-0.728
102	115.549	0.354	17.179	8.284	-0.785
105	115.550	0.341	17.177	8.280	-0.771
107	102.389	16.127	28.083	25.136	-27.065
109	102.389	16.127	28.083	25.136	-27.065
111	115.521	1.657	17.162	8.234	-3.502
113	108.072	0.894	22.874	17.495	-2.076
115	105.677	12.211	25.360	20.931	-20.523
119	105.677	12.211	25.360	20.931	-20.523
123	115.526	1.359	17.169	8.255	-2.851
127	109.939	0.786	21.453	15.200	-1.781
131	56.168	16.007	21.214	21.828	-26.774
133	56.168	16.007	21.214	21.828	-26.774
135	69.300	1.538	10.293	4.927	-3.211
137	61.851	0.774	16.005	14.188	-1.784

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	161.773	0.419	24.042	11.576	-1.019

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
202	138.657	0.447	20.617	9.947	-0.965
204	138.659	0.414	20.613	9.938	-0.931
206	128.132	13.021	29.335	23.416	-21.943
208	128.132	13.021	29.335	23.416	-21.943
214	117.598	25.738	38.069	36.926	-43.070
218	117.598	25.738	38.069	36.926	-43.070
222	138.614	2.354	20.598	9.889	-4.894
224	127.973	1.263	28.758	23.119	-2.856
226	82.936	25.593	32.911	34.429	-42.794
228	82.936	25.593	32.911	34.429	-42.794
230	103.952	2.209	15.440	7.393	-4.618
232	93.311	1.119	23.600	20.623	-2.580

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 4.410m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.091m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 2.100 m Governing Load Case : # 101

Width (W_2) = 2.100 m Governing Load Case : # 101

Depth (D_2) = 0.400 m Governing Load Case : # 214

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 4.410 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 42.335 kN

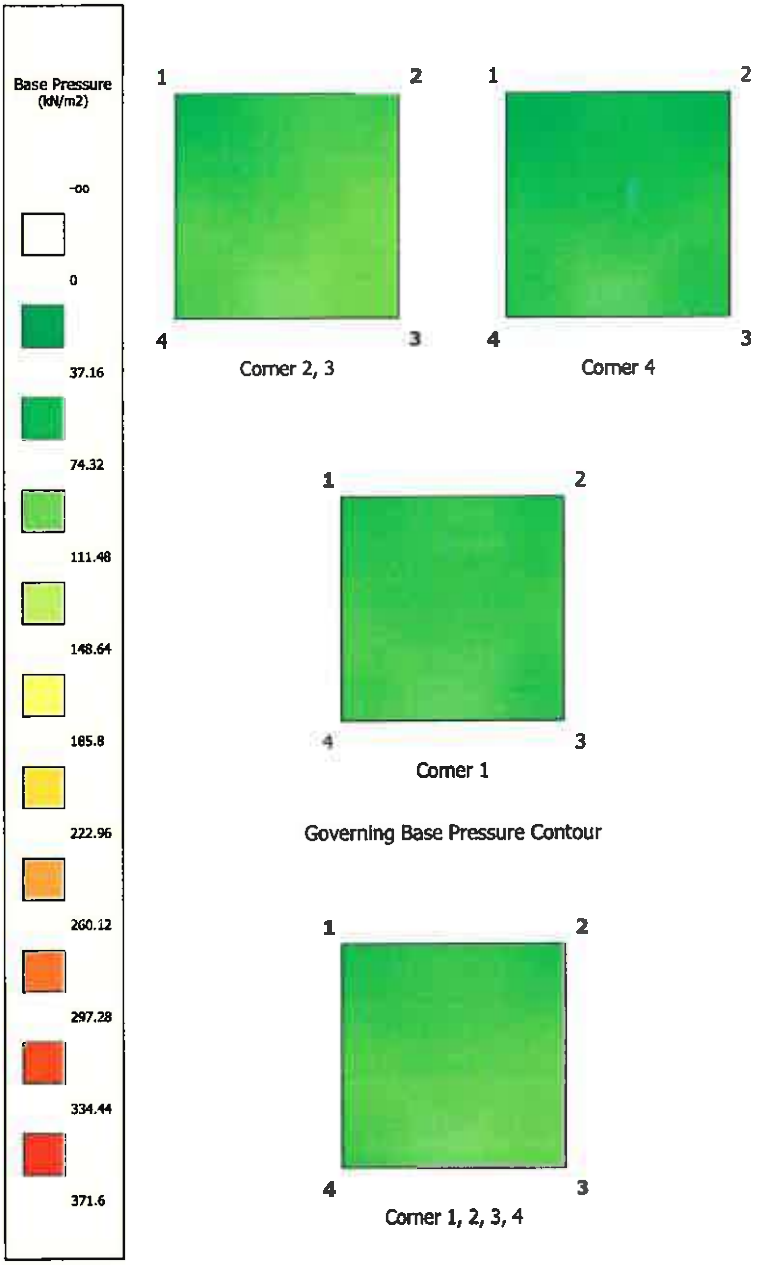
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 60.910 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



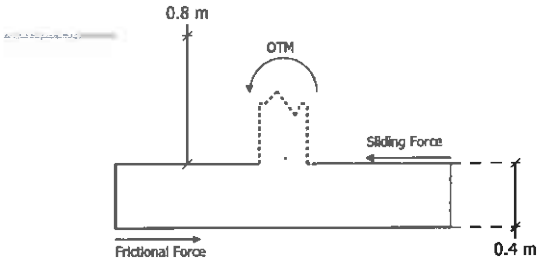
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**39.2572**	40.3557	59.9702	58.8718	0.000
107	1.3518	**44.7803**	91.9060	48.4774	0.000
107	1.3518	44.7803	**91.9060**	48.4774	0.000
113	29.0784	32.2311	66.7566	**63.6039**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**39.2572**	40.3557	59.9702	58.8718
107	1.3518	**44.7803**	91.9060	48.4774
107	1.3518	44.7803	**91.9060**	48.4774
113	29.0784	32.2311	66.7565	**63.6039**

Stability Check



	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
101	423.823	7.390	10.449	1.500	15.177	273.404	1.500
102	358.261	7.387	10.445	1.500	15.158	250.101	1.500
105	372.674	7.388	10.446	1.500	15.163	255.547	1.500
107	7.396	4.247	5.208	1.500	5.937	6.503	1.500
109	7.396	4.247	5.208	1.500	5.937	6.503	1.500
111	76.555	7.393	10.407	1.500	15.213	55.638	1.500
113	137.113	5.358	7.572	1.500	8.327	92.031	1.500
115	9.923	4.778	6.088	1.500	7.059	8.714	1.500
119	9.923	4.778	6.088	1.500	7.059	8.714	1.500
123	93.367	7.390	10.419	1.500	15.190	68.261	1.500
127	157.239	5.764	8.145	1.500	9.412	107.769	1.500
131	4.893	3.692	4.168	1.500	4.678	4.311	1.500
133	4.893	3.692	4.168	1.500	4.678	4.311	1.500
135	55.893	8.350	11.679	1.500	17.203	40.985	1.500
137	105.441	5.100	7.204	1.500	7.177	71.152	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : **131**

Governing Disturbing Force : **16.007kN**

Governing Restoring Force : **78.328kN**

Minimum Sliding Ratio for the Critical Load Case : **4.893**

Critical Load Case for Overturning about X-Direction : **131**

Governing Overturning Moment : **30.314kNm**

Governing Resisting Moment : **141.798kNm**

Minimum Overturning Ratio for the Critical Load Case : **4.678**

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 131
Governing Disturbing Force : 21.214kN
Governing Restoring Force : 78.328kN
Minimum Sliding Ratio for the Critical Load Case : 3.692
Critical Load Case for Overturning about Z-Direction : 131
Governing Overturning Moment : -33.176kNm
Governing Resisting Moment : 143.023kNm
Minimum Overturning Ratio for the Critical Load Case : 4.311

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

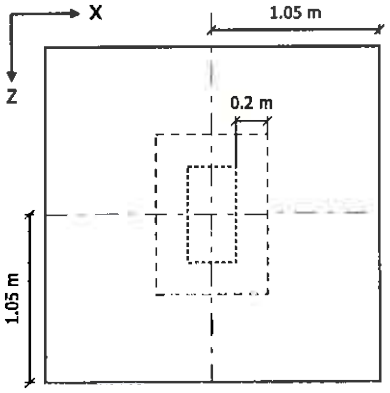
Critical Load Case for Sliding along Resultant Direction : 131
Governing Disturbing Force : 26.575kN
Governing Restoring Force : 110.773kN
Minimum Sliding Ratio for the Critical Load Case : 4.168

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m
Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m
For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until 0.75XV_c ≥ Punching Shear Force

Punching Shear Force, Vu = 264.088kN, Load Case # 201

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.176m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2146.802kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 3398.656kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2146.802kN

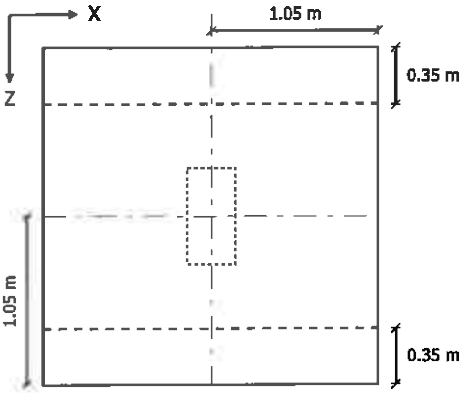
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1610.102kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = **709.743kN**

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = **0.406m**

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

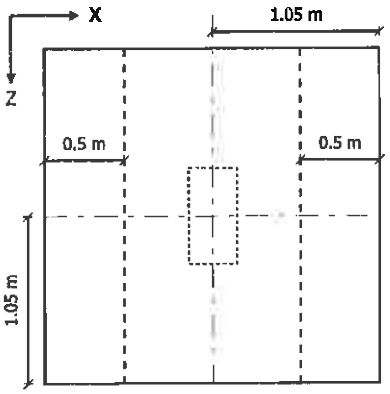
From above calculations, 0.75 X V_c = 532.308 kN

Critical load case for V_{ux} is # 214 $V_{ux} = V_{ux}|_{z=D_z}$ = 69.539 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 709.743 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{zd}$ = 0.556 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

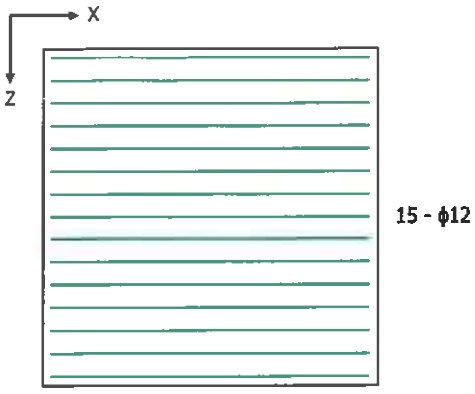
From above calculations, 0.75 X V_c = 532.308 kN

Critical load case for V_{uz} is # 214 $V_{uz} = V_{uz}|_{z=D_z}$ = 93.228 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 214

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00164
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.200 m
Ultimate moment,	$M_u\big\vert_{z=D_x}$ =	68.021 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	75.579 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00073
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00063
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	ρ x W x d_{eff} =	1380.565 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 50.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c'}} =$	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) - C_{cover} =$	0.850	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}} =$	15	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1696.448	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.344	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)} =$	0.00235	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

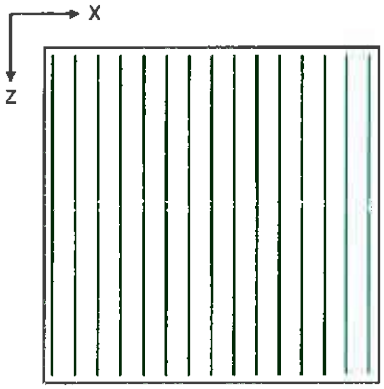
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 50.000mm

Provided Steel Area / Required Steel Area = 1.229

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



15 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 214

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.332 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal} =$	0.02488
From ACI Cl.7.12.2, ρ_{min} =		0.00159
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{(0.85 \times F_c')} =$	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z =	$0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ =	1.350	m
Ultimate moment,	$M_u\big\vert_{x=D_z}$ =	48.055	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	53.394	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00055	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00046	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	ρ x W x d_{eff} =	1335.422	mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 50.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{\omega_y \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) - C_{cover}$ =	0.700	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	15	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

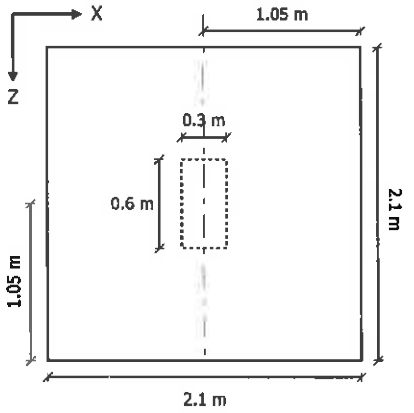
Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1696.448	mm2
d_{eff} =	D - C_{cover} - 1.5 X (dia. of one bar) =	0.332	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00243	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

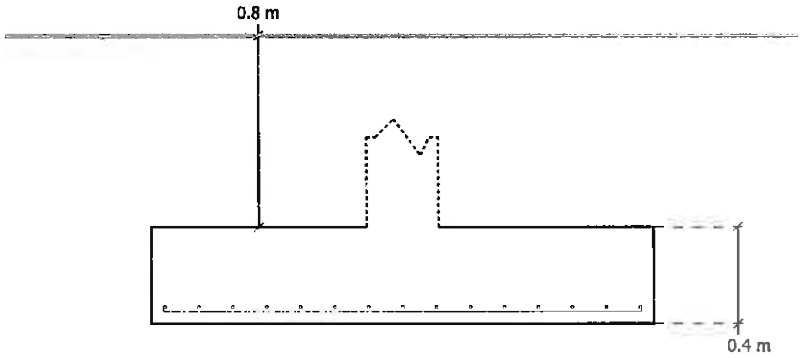
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 50.000mm

Provided Steel Area / Required Steel Area = 1.270

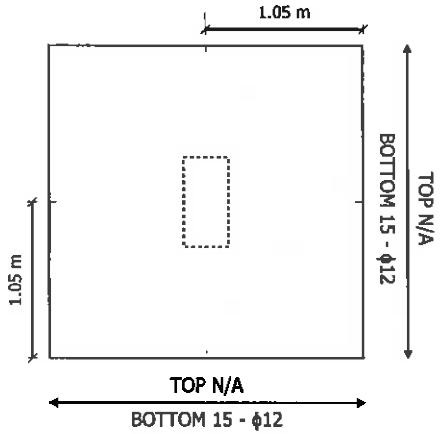
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

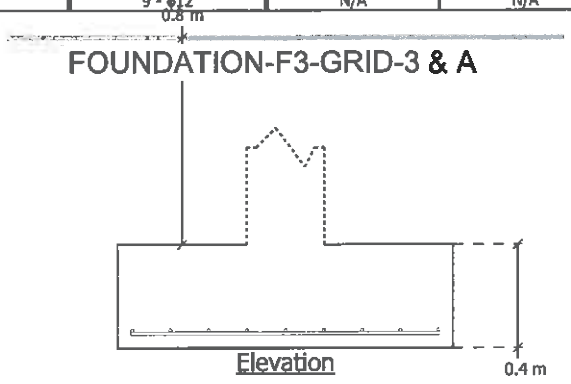
	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	2.1 m	2.1 m	0.4 m	N/A	N/A	15 - ϕ12	15 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

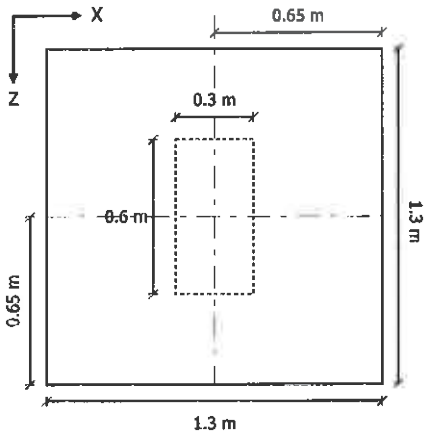
Isolated Footing Design(ACI 318-08)

Footing No.	Group ID	Foundation Geometry		
		Length	Width	Thickness
38	1	1.300m	1.300m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
38	9 - ϕ12	9 - ϕ12	N/A	N/A	N/A	N/A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1300.000mm
Footing Width - Z (Fw) : 1300.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.300m

Column Width - Z (B_{col}) : 0.600m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 50.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.580
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.300m

Initial Width (W_o) = 1.300m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
101	SL:1.000 x DL	1.00	1.00	1.00
102	SL:1.000 x DL+1.000 x LL	1.00	1.00	1.00
105	SL:1.000 x DL+0.750 x LL	1.00	1.00	1.00
107	SL:1.000 x DL+1.000 x WLX	1.00	1.00	1.00
109	SL:1.000 x DL+1.000 x WLZ	1.00	1.00	1.00
111	SL:1.000 x DL+0.700 x ELX	1.00	1.00	1.00
113	SL:1.000 x DL+0.700 x ELZ	1.00	1.00	1.00
115	SL:1.000 x DL+0.750 x LL+0.750 x WLX	1.00	1.00	1.00
119	SL:1.000 x DL+0.750 x LL+0.750 x WLZ	1.00	1.00	1.00
123	SL:1.000 x DL+0.750 x LL+0.525 x ELX	1.00	1.00	1.00
127	SL:1.000 x DL+0.750 x LL+0.525 x ELZ	1.00	1.00	1.00
131	SL:0.600 x DL+1.000 x WLX	1.00	1.00	0.60
133	SL:0.600 x DL+1.000 x WLZ	1.00	1.00	0.60
135	SL:0.600 x DL+0.700 x ELX	1.00	1.00	0.60
137	SL:0.600 x DL+0.700 x ELZ	1.00	1.00	0.60

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
201	UL:1.400 x DL	1.00	1.00	1.40
202	UL:1.200 x DL+1.600 x LL	1.00	1.00	1.20
204	UL:1.200 x DL+1.000 x LL	1.00	1.00	1.20
206	UL:1.200 x DL+0.800 x WLX	1.00	1.00	1.20
208	UL:1.200 x DL+0.800 x WLZ	1.00	1.00	1.20
214	UL:1.200 x DL+1.000 x LL+1.600 x WLX	1.00	1.00	1.20
218	UL:1.200 x DL+1.000 x LL+1.600 x WLZ	1.00	1.00	1.20
222	UL:1.200 x DL+1.000 x LL+1.000 x ELX	1.00	1.00	1.20
224	UL:1.200 x DL+1.000 x LL+1.000 x ELZ	1.00	1.00	1.20
226	UL:0.900 x DL+1.600 x WLX	1.00	1.00	0.90
228	UL:0.900 x DL+1.600 x WLZ	1.00	1.00	0.90
230	UL:0.900 x DL+1.000 x ELX	1.00	1.00	0.90
232	UL:0.900 x DL+1.000 x ELZ	1.00	1.00	0.90

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	140.310	-7.919	7.654	3.584	4.709
102	144.230	-8.122	7.671	3.587	4.909
105	143.250	-8.071	7.667	3.586	4.859
107	151.486	-6.919	18.041	25.160	2.894
109	151.486	-6.919	18.041	25.160	2.894
111	134.054	-4.340	7.637	3.534	0.739
113	149.030	-8.021	13.125	15.391	4.774
115	151.632	-7.321	15.457	19.769	3.498
119	151.632	-7.321	15.457	19.769	3.498
123	138.558	-5.387	7.654	3.549	1.882
127	149.790	-8.147	11.769	12.442	4.908
131	95.362	-3.752	14.980	23.727	1.011
133	95.362	-3.752	14.980	23.727	1.011
135	77.930	-1.173	4.575	2.100	-1.145
137	92.906	-4.853	10.063	13.957	2.891

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	196.434	-11.086	10.716	5.017	6.592

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
202	174.644	-9.827	9.212	4.306	5.971
204	172.292	-9.705	9.202	4.304	5.851
206	177.312	-8.703	17.495	21.562	4.199
208	177.312	-8.703	17.495	21.562	4.199
214	190.173	-8.106	25.821	38.826	2.947
218	190.173	-8.106	25.821	38.826	2.947
222	163.354	-4.593	9.177	4.233	0.180
224	184.749	-9.851	17.016	21.171	5.944
226	144.160	-5.528	23.508	37.748	1.334
228	144.160	-5.528	23.508	37.748	1.334
230	117.342	-2.015	6.864	3.154	-1.433
232	138.736	-7.272	14.704	20.092	4.331

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 1.690m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 0.982m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.300 m Governing Load Case : # 101

Width (W_2) = 1.300 m Governing Load Case : # 101

Depth (D_2) = 0.400 m Governing Load Case : # 214

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 1.690 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 16.223 kN

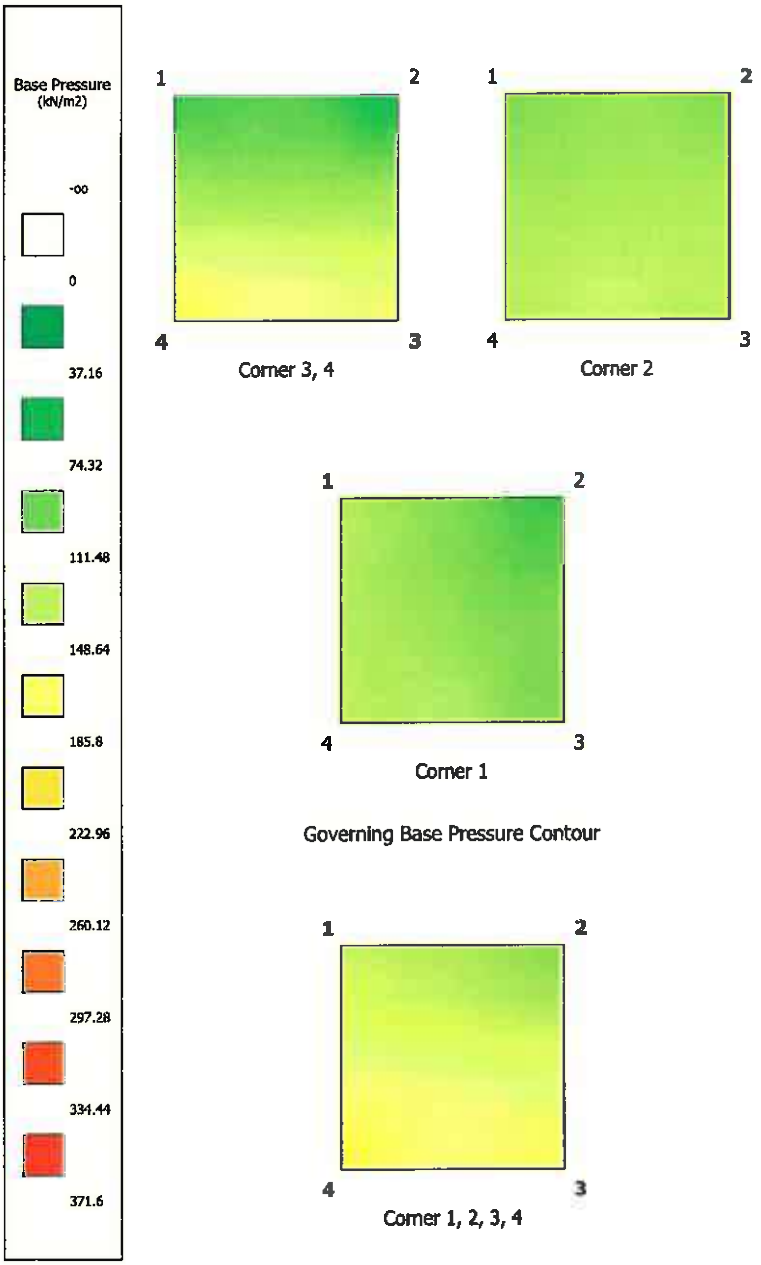
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 21.743 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

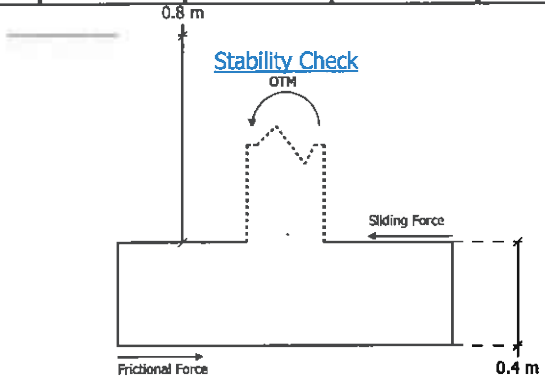


Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
102	**111.9107**	67.3530	103.7064	148.2642	0.000
111	90.5527	**77.0341**	113.0218	126.5404	0.000
107	39.1421	8.2173	**185.0618**	215.9865	0.000
107	39.1421	8.2173	185.0618	**215.9865**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
102	**111.9107**	67.3530	103.7064	148.2642
111	90.5527	**77.0341**	113.0218	126.5404
107	39.1421	8.2173	**185.0618**	215.9865
107	39.1421	8.2173	185.0618	**215.9865**



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
101	13.058	13.509	13.278	1.500	17.437	14.873	1.500
102	13.011	13.776	13.377	1.500	17.793	14.672	1.500
105	13.023	13.710	13.353	1.500	17.704	14.721	1.500
107	15.881	6.091	8.042	1.500	3.803	21.973	1.500
109	15.881	6.091	8.042	1.500	3.803	21.973	1.500
111	22.988	13.065	16.063	1.500	16.970	45.687	1.500
113	13.522	8.264	9.972	1.500	5.889	15.385	1.500
115	15.020	7.114	9.093	1.500	4.749	19.373	1.500
119	15.020	7.114	9.093	1.500	4.749	19.373	1.500
123	19.006	13.377	15.470	1.500	17.357	28.739	1.500
127	13.366	9.253	10.759	1.500	7.116	15.098	1.500
131	19.267	4.826	6.620	1.500	2.726	32.561	1.500
133	19.267	4.826	6.620	1.500	2.726	32.561	1.500
135	53.019	13.590	18.617	1.500	17.728	104.272	1.500
137	14.601	7.042	8.970	1.500	4.416	16.592	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : **102**
Governing Disturbing Force : -8.122kN
Governing Restoring Force : 105.674kN
Minimum Sliding Ratio for the Critical Load Case : 13.011
Critical Load Case for Overturning about X-Direction : 131
Governing Overturning Moment : 29.719kNm
Governing Resisting Moment : 81.009kNm
Minimum Overturning Ratio for the Critical Load Case : 2.726

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 131
Governing Disturbing Force : 14.980kN
Governing Restoring Force : 72.286kN
Minimum Sliding Ratio for the Critical Load Case : 4.826
Critical Load Case for Overturning about Z-Direction : 102
Governing Overturning Moment : 8.158kNm
Governing Resisting Moment : 119.689kNm
Minimum Overturning Ratio for the Critical Load Case : 14.672

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

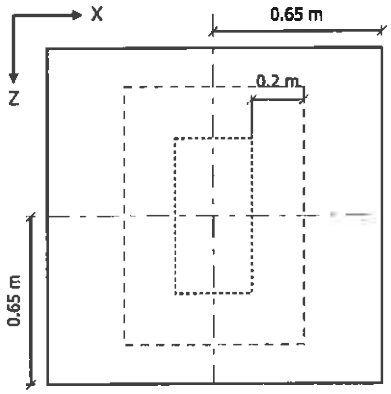
Critical Load Case for Sliding along Resultant Direction : 131
Governing Disturbing Force : 15.442kN
Governing Restoring Force : 102.228kN
Minimum Sliding Ratio for the Critical Load Case : 6.620

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m
Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m
For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until 0.75XV_c ≥ Punching Shear Force

Punching Shear Force, Vu = 159.805kN, Load Case # 201

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.176m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f'_c} \times b_o \times d_{eff}$ 2146.802kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \sqrt{f'_c} \times b_o \times d_{eff}$ 3398.656kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f'_c} \times b_o \times d_{eff}$ 2146.802kN

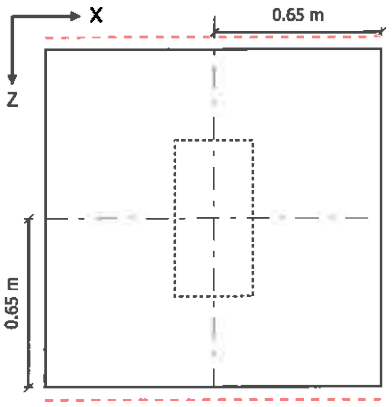
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1610.102kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 439.365kN

Distance along X to design for shear, D_x $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.006m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

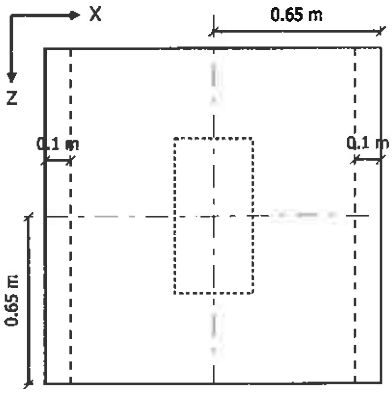
From above calculations, 0.75 X V_c = 329.524 kN

Critical load case for V_{ux} is # 214 $V_{ux} = V_{ux}|_{z=D_z}$ = 2.091 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 439.365 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{zd}$ = 0.156 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

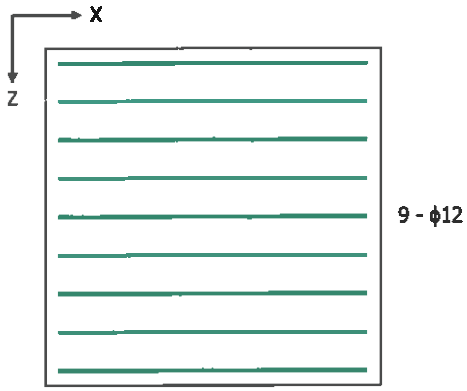
From above calculations, 0.75 X V_c = 329.524 kN

Critical load case for V_{uz} is # 201 $V_{uz} = V_{uz}|_{z=D_z}$ = 35.325 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 201

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f'_c \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00177
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.500 m
Ultimate moment,	$M_u\big\|_{z=D_z}$ =	27.636 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	30.707 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00048
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00041
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	918.298 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 50.000mm

Selected spacing (S) = 148.500mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 145mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.450 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 9

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1017.869 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00228

From ACI Cl.7.6.1, minimum req'd clear distance between bars

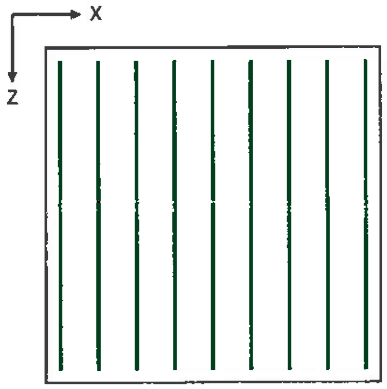
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 50.000mm

Provided Steel Area / Required Steel Area = 1.108

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



9 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 214

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f'_c \times \left(\frac{\quad}{\quad + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00174

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F'_c)}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_x = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.950 m

Ultimate moment, $M_u|_{x=D_x}$ = 19.948 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 22.165 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00037

(Based on gross depth) ρ x d_{eff} / Depth = 0.00031

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 902.299 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 50.000mm

Selected spacing (S) = 150.000mm

$S_{min} <= S <= S_{max}$ and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.300 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 9

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1017.869 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

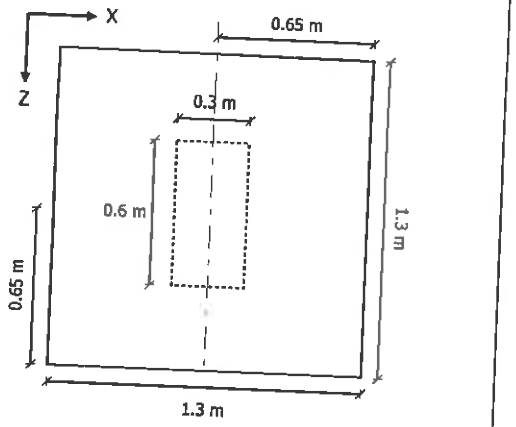
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00236

From ACI Cl.7.6.1, minimum req'd clear distance between bars

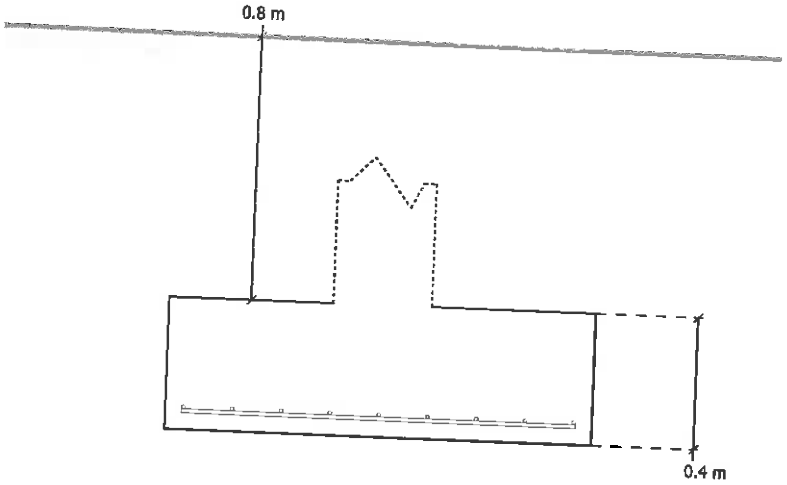
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 50.000mm

Provided Steel Area / Required Steel Area = 1.128

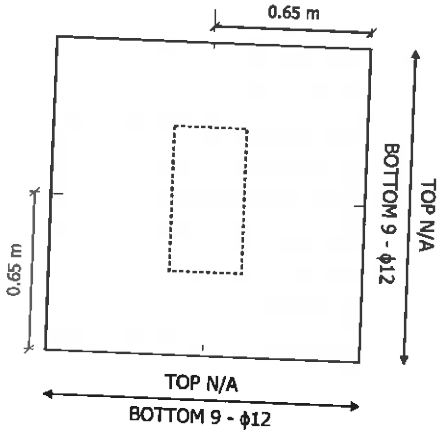
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.3 m	1.3 m	0.4 m	N/A	N/A	9 - ϕ12	9 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet